| 3rd Semester | Integrated Professional Course (IPC)<br>OPERATING SYSTEM | M25BCS301 |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic programming skills | Understanding of programming concepts and experience in a programming language such as C, C++, Java, or Python |
| 2 | Computer architecture | Basic understanding of computer organization and architecture, including concepts like memory hierarchy, CPU operations, and input/output systems. |
| 3 | Data structures and algorithms | Familiarity with fundamental data structures (arrays, linked lists, trees, etc.) and algorithms (sorting, searching, etc.). |
| 4 | Computer Organization | Knowledge of how hardware components interact at a low level is helpful. This includes concepts like machine instructions, addressing modes, memory management, and I/O operation |
| 5 | Databases | Basic understanding of databases and file management systems is helpful as operating systems often interact with databases for storing and retrieving data. |

**2. Syllabus**

| OPERATING SYSTEM<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS301** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **48:0:32:30:0 = 110 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **04** | Exam Hours | **03** |

**Course objectives:**

This course will enable students to:

1. Basic Understanding of Computer System Structure and Operating Systems Structure
2. Analyse the main tasks carried out by the operating systems Process and thread management, CPU scheduling algorithms, memory management and deadlocks.
3. To demonstrate different APIs/Commands related to processor, memory, storage and file system management.

**Module -1**

**Introduction to operating systems:** What operating systems do; Computer System organization; Computer System architecture; Operating System operations(dual-mode and multi-mode);computing environments; **System structures:** Operating system services; User - Operating System interface; System calls; Types of system calls; operating system structures.

**Textbook 1- chapter 1(1.1,1.2,1.3,1.5,1.11) chapter 2 (2.1,2.2,2.3,2.4,2.7) 8 Hours**

**Module -2**

**Process managemen**t: Process Concept; Process Scheduling; Operation on Process; Inter-Process Communication. **Multithreaded programming**: Overview; Multicore programming, multithreading models, thread libraries, threading issues. **Process scheduling** - Basic Concepts, CPU I/O Burst Cycle; CPU Scheduler – Pre-emptive Scheduling, Dispatcher; Scheduling Criteria; Scheduling Algorithms – FCFS, SJF, Round-Robin, Priority.

**Textbook 1-chapter 3(3.1,3.2,3.3,3.4) chapter 4(4.1,4.2,4.3,4.4,4.6) chapter 6(6.1,6.2,6.3) 8 Hours**

**Module -3**

**Process Synchronization:** Synchronization: The critical section problem; Peterson's solution; Synchronization hardware; Mutex locks; Semaphores; Classical problems of synchronization; **Deadlocks**: System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock.

**Textbook 1-chapter 5(5.1,5.2,5.3,5.4,5.5,5.6,5.7) chapter 7(7.1-7.7) 8 Hours**

**Module -4**

**Memory Management**: Background; Swapping; Contiguous memory allocation; Segmentation; Paging; Structure of page table; **Virtual Memory Management**: Background; Demand paging; Page replacement; Allocation of frames; Thrashing.

**Textbook 1-chapter 8(8.1-8.6) chapter 9(9.1,9.2,9.4,9.5,9.6) 8 Hours**

**Module -5**

**File System**: File concept; Access methods; Directory and Disk structure; File system mounting; File sharing; **Storage management:** overview of Mass storage structures; Disk structure; Disk attachment; Disk

scheduling;

**Textbook 1-chapter 11(11.1-11.5) chapter 10(10.1-10.4)** **10 Hours**

**TEXTBOOKS:**

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, "Operating System Concepts", 9th edition, Wiley-India, 2018
2. D.M Dhamdhere, "Operating Systems: A Concept Based Approach" 3rd Ed, McGraw- Hill, 2013

**REFERENCE BOOKS:**

1. P.C.P. Bhatt, "An Introduction to Operating Systems: Concepts and Practice", 4th Edition, PHI(EEE), 2014.
2. William Stallings, "Operating Systems: Internals and Design Principles", 6th Edition, Pearson.

**VIDEO LINKS:**

1. https://nptel.ac.in/courses/10610521
2. https://www.geeksforgeeks.org/operating-systems/

**PRACTICAL COMPONENT**

**Description:** Implement all the program in c programming language.

| | |
|---|---|
| 1. | Your company is developing software where a parent process needs to spawn multiple child processes to handle independent tasks. The child process should replace its execution image with another program and report back to the parent. (Hint: Process Creation & System Calls). |
| 2. | A system has multiple jobs arriving at different times, and you are responsible for implementing a scheduler to minimize waiting time and turnaround time. (Hint: CPU Scheduling (FCFS & Round Robin)). |
| 3. | A factory has multiple producers placing items in a buffer and multiple consumers taking items from it. You need to ensure proper synchronization to avoid race conditions and buffer overflow/underflow. (Hint: Producer-Consumer Problem (Semaphores)). |
| 4. | Two processes, one for writing data and another for reading it, need to communicate using a named pipe (FIFO). (Hint: Interprocess Communication (Reader & Writer using FIFO)). |
| 5. | A banking system wants to allocate resources to multiple clients while ensuring that deadlock is avoided. (Hint: Banker's Algorithm (Deadlock Avoidance)). |
| 6. | A system has contiguous blocks of memory and multiple processes requesting memory of varying sizes. You need to allocate memory efficiently. (Hint: Contiguous Memory Allocation (Worst Fit & Best Fit)). |
| 7. | A system with limited memory uses paging. You must design a page replacement strategy to minimize page faults. (Hint: Page Replacement Algorithms (FIFO & LRU)). |
| 8. | A file system is to be implemented with different directory structures to manage files efficiently. (Hint: File Organization Techniques (Single & Two-Level Directory)). |
| 9. | A hard disk scheduling system needs to minimize seek time. The head is at a given position, and requests arrive in random order. (Hint: Disk Scheduling (SCAN Algorithm)). |

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of operating system concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |
| 5 | Problem-Based Learning (PBL) | Implement PBL to enhance analytical skills and practical application of competencies |
| 6 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |

4. **Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Integrated Professional Course (IPC)**

| Components | | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| **Theory (A)** | Internal Assessment - Tests | 3 | 60% | 15 |
| | Term Work - TW | 2 | 40% | 10 |
| | **Total Marks** | | **100%** | **25** |
| **Laboratory (B)** | Record Writing | Continuous | 60% | 15 |
| | Test at the end of the semester | 1 | 40% | 10 |
| | **Total Marks** | | **100%** | **25** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks + Term Work marks

**B** = Average Laboratory marks

*** Assessment / Case Studies / Practices**

1. **Assignment and Quiz**
2. **Mini-Project**
3. **Group Presentation**

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. The question paper may include at least one question from the laboratory component.
5. Marks scored will be proportionally scaled down to 50 marks Course Outcomes (COs) and Mapping with POs/ PSOs

5. **Course Outcomes (COs) and Mapping with POs and PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS301.1** | Understand and apply the fundamental concepts and principles of operating systems. |
| **M25BCS301.2** | Analyse various inter-process communication, multiprogramming mechanisms and apply different process scheduling algorithms. |
| **M25BCS301.3** | Examine multiple mechanisms for managing deadlock situations and Implement both software and hardware solutions to address the critical-section problem |
| **M25BCS301.4** | Determine and evaluate memory management techniques. |
| **M25BCS301.5** | Interpret the structure of file systems and the organization of secondary storage devices |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS301.1** | 3 | - | - | - | | - | - | - | - | - | - | 2 | - |
| **M25BCS301.2** | - | 3 | 3 | - | 3 | - | - | - | - | - | - | - | 2 |
| **M25BCS301.3** | - | - | 3 | - | 3 | - | - | - | - | - | - | - | 2 |
| **M25BCS301.4** | 3 | - | 3 | 3 | - | - | - | - | - | - | - | 2 | 2 |
| **M25BCS301.5** | - | - | 3 | - | 3 | - | - | - | - | - | - | - | - |
| **M25BCS301** | **3** | **3** | **3** | **3** | **3** | - | - | - | - | - | - | **2** | **2** |

6. **Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4**: Quality Education | Using OS simulations and open-source tools helps improve CSE education accessibility. |
| 2 | **SDG 9**: Industry, Innovation, and Infrastructure | OS are foundational for cloud computing, IoT, and many other industries. |
| 3 | **SDG 12**: Responsible Consumption and Production | OS handles optimal resource usage, supporting efficiency in digital devices and memory utilization |

| 3rd Semester | Integrated Professional Course (IPC)<br>OOPS WITH JAVA | M25BCS302 |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic Computer Skills | Basic computer skills, such as saving files in multiple versions and formats. |
| 2 | Programming Fundamentals | Familiar with general coding concepts like variables, basic data types, Conditional Statements, Looping, Functions, creation of source file, compilation process, program execution techniques. |
| 3 | Multi-Process Execution Programming | Familiar with the way to execute multiple tasks simultaneously by creating multiple processes. |
| 4 | Basic Object Orientation Concepts | Basic of four basic principles: encapsulation, inheritance, polymorphism, and abstraction. Where these four OOP principles can be used enable to create objects and collaborate to create powerful applications too. |
| 5 | Programming basic tools | Familiar with Programming tools like assemblers, compilers, linkers translate, flowchart, algorithms which can be used to form a program from a human write-able and readable source language into the bits and bytes that can be executed by a computer. |

**2. Syllabus**

| OOPS WITH JAVA<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS302** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **48:0:32:30:0 = 110 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **04** | Exam Hours | **03** |

**Course Objectives:**

1. To learn primitive constructs JAVA programming language.
2. To understand Object Oriented Programming Features of JAVA.
3. To gain knowledge on: packages, multithreaded programming and exceptions.

**Module -1**

**Introduction to Java**: Java Buzz Words, Java's Magic: Bytecode, Two Paradigms of OOPS, Principles of OOPS, Lexical Issues, Use Block of Code, First Java Program.
**Variables in Java**: Data Types, Literals Constants, Variables, Type Conversion.
**Arrays**: One-dimensional array, multi-dimensional array, alternative array declaration, Java Strings.
**Operators**: Arithmetic Operators, Relational Operators, Logical Operators, Bitwise Operators, The Assignment Operator, The Conditional Operator, Operator Precedence.
**Selected Topics from Textbook 1: Chapter 1, 2, 3, 4** **8 Hours**

**Module -2**

**Control Statements:** Selection Statements, Iteration Statements, Jump Statements.
**Introduction to Classes**: Class Fundamentals, Declaring Objects, Object Reference Variables, Garbage Collection.
**Methods**: Introducing Class Methods, Constructors, This Keyword, Example Java Programs with Class and Methods.
**Selected Topics from Textbook 1: Chapter 5, 6, 7** **8 Hours**

**Module -3**

**Polymorphism:** Overloading Methods, Overloading Constructors, Using Objects as Parameters and Argument Passing, Returning Objects.
**Access Control:** Introduction to Access specifier, Understanding Static Members & Methods, Nested and Inner Classes.
**Inheritance:** Inheritance Basics, Member access and inheritance, using super keyword, Constructors call in Multilevel Hierarchy, Method Overriding, Abstract Classes, Final and Inheritance.
**Selected Topics from Textbook 1: Chapter 7, 8** **8 Hours**

**Module -4**

**Packages:** Packages, defining a Package, Access protection, Importing Packages.
**Interfaces:** Defining Interfaces, implementing an interface, Access Implementations through interface references, Nested Interface.
**Exceptions-Handling:** Fundamentals of Exception-Handling Fundamentals, Exception Types, Using different types of Exception clauses, Nested try statements, Creating Your Own Exception Subclasses.
**Selected Topics from Textbook 1: Chapter 9, 10** **8 Hours**

**Module -5**

**Multithreaded Programming:** The Java Thread Model, The Main Thread, creating a user thread, Creating Multiple Threads, Synchronization, Inter-thread Communication, Deadlocks, Suspending, Resuming, and Stopping Threads, Obtaining a Thread's State.
**Selected Topics from Textbook 1:Chapter 11** **8 Hours**

**TEXTBOOKS:**

1. Herbert Schildt, "Java: The Complete Reference", 8thvEdition, November 2012, McGraw-Hill Edition 2011, ISBN: 978-1-25-900246-5.
2. E Balagurusamy, "Programming with Java A Primer", 4th Edition, Mar-2010, Tata McGraw Hill Education, ISBN: 978-0-07-014169-8.

**REFERENCE BOOKS:**

1. M P Bhave and S A Patekar, "Programming with JAVA", 5th Edition, 2017, Pushp Print Services, ISBN:978-81-317-2080-6.
2. Bruce Eckel, "Thinking in Java", Fourth Edition, , Prentice Hall, 2006 (https://sd.blackball.lv/library/thinking_in_java_4th_edition.pdf)

**VIDEO LINKS:**

1. https://onlinecourses.nptel.ac.in/noc25_cs110/preview

**PRACTICAL COMPONENT**

| | |
|---|---|
| 1. | A school wants to store student grades for multiple classes using two-dimensional arrays. Develop a Java program that demonstrates different ways to declare and initialize two-dimensional arrays to record these grades. Show how to handle both fixed-size and variable-sized arrays for different classes and subjects. |
| 2. | During the annual interschool quiz, the coordinator is required to announce the names of the registered participants from each school. The names of the participants are displayed for attendance purposes. Write a Java program to print the list of student names using a for-each loop. |
| 3. | Create a class named Student to represent the details of a student, including USN, Name, IA1_Marks, IA2_Marks, IA3_Marks, and Avg_Marks. Write a member function, computeAvg(m1, m2, m3), to calculate the average of the three IA marks and store it in Avg_Marks. Develop a suitable main method to demonstrate the creation of Student objects, computation of the average, and display the details for each student. Create a class named Student to represent the details of a student, including USN, Name, IA1_Marks, IA2_Marks, IA3_Marks, and Avg_Marks. Write a member function, computeAvg(m1, m2, m3), to calculate the average of the three IA marks and store it in Avg_Marks. Develop a suitable main method to demonstrate the creation of Student objects, computation of the average, and display the details for each student. |
| 4. | In a software company, employees are categorized into different roles such as Manager, Developer, and Programmer. At the time of the annual appraisal, the company decides to provide different salary hikes based on job titles: Managers get a 40% raise, Developers get a 25% raise, and Programmers receive a 10% raise. The HR department needs a program to manage employee information and calculate new salaries according to these criteria. |
| 5. | In a vehicle manufacturing company, different types of vehicles go through several stages of assembly. The base class Vehicle represents the initial assembly, the Car class adds body and interiors, and the ElectricCar class finalizes the electric vehicle-specific components. When a new electric car is assembled, constructors for each stage are called in order, ensuring each level of the process is properly initialized. Consider a vehicle manufacturing process with multiple assembly stages:<br>• Create a Java program with a multi-level class hierarchy:<br>o The base class should be Vehicle.<br>o Car should inherit from vehicle.<br>o Electric car should inherit from car |
| 6. | Design a Java program with an interface Sortable which declares a method sort() for sorting an array of integers in ascending order. Implement two classes, Bubble Sort and SelectionSort, that implement the Sortable interface and provide their specific versions of the sort() method. Demonstrate the sorting of an integer array using both sorting techniques in the main method. |
| 7. | Demonstrate how to create a package named MyPack in Java with a class called MyClass that contains a method MyMethod(). Then, create a separate file New.java that imports the |

| | |
|---|---|
| | MyPackpackage, creates an object of MyClass, and calls the MyMethod() method inside the main method. Provide the complete code for both files and explain the process of creating, importing, and using packages in Java. |
| 8. | A text validation system for a language learning app needs to ensure that user inputs contain vowels to support pronunciation exercises. If the input string does not contain any vowels, the system should throw an exception to prompt the user to enter a valid word containing vowels. Develop a Java program with a method that takes a string as input and throws an exception if the string does not contain any vowels. |
| 9. | In a text processing application, different tasks need to be handled concurrently for efficiency. One thread performs string concatenation operations while another thread converts the resulting string to uppercase letters. Write a Java program to create a child thread by implementing the Runnable interface. The child thread should perform string concatenation, while the main thread converts the final concatenated string to uppercase. Demonstrate how both threads work concurrently in the program. |
| 10. | Develop a Java program with three classes: Storage, Counter, and Printer. The Storage class should hold an integer value. The Counter class should create a thread that counts upwards from 0, storing each count value in the Storage class. The Printer class should create a thread that continuously reads the value from the Storage and prints it. Demonstrate how these threads operate concurrently to share and display the count in real time. |

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies |
| 2 | Image/Video/Animation | Incorporate visual aids like image/videos/animations to enhance understanding of programming constructs. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |
| 5 | Programming-Based Learning (PBL) | Implement PBL to enhance analytical skills and practical application of competencies. |
| 6 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 7 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies. |
| 8 | Programming Assignments | Assign programming tasks to reinforce practical skills associated with competencies. |

4. **Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Integrated Professional Course (IPC)**

| Components | | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| **Theory (A)** | Internal Assessment - Tests | 3 | 60% | 15 |
| | Term Work - TW | 2 | 40% | 10 |
| | **Total Marks** | | **100%** | **25** |
| **Laboratory (B)** | Record Writing | Continuous | 60% | 15 |
| | Test at the end of the semester | 1 | 40% | 10 |
| | **Total Marks** | | **100%** | **25** |

**Final CIE Marks = (A) + (B)**

**A = Average of best two Test marks + Average of two Term Work marks**

**B = Average Laboratory marks**

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.

3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS302.1** | Understand and apply the basic programming constructs. |
| **M25BCS302.2** | Apply the structure of classes and methods in Java programming environment. |
| **M25BCS302.3** | Analyse the different programming constructs of Java and its effectiveness in improving the efficiency of Java programs. |
| **M25BCS302.4** | Implement appropriate Java programming constructs to solve real-world problem sample scenarios. |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS302.1** | 3 | - | - | - | - | - | - | - | - | - | - | 2 | - |
| **M25BCS302.2** | 3 | - | - | - | 3 | - | - | - | - | - | - | - | 2 |
| **M25BCS302.3** | - | 3 | - | - | 3 | - | - | - | - | - | - | 2 | 2 |
| **M25BCS302.4** | - | - | 3 | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS302** | **3** | **3** | **3** | - | **3** | - | - | - | - | - | - | **2** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 3**: Good Health and Well-Being | Java is used in **medical software** (patient records, diagnostic tools). Promotes **safe and secure systems** through encapsulation and validation mechanisms. (SDG 3). |
| 2 | **SDG 9:** Industry, Innovation and Infrastructure, | Java is widely used in **enterprise applications** (banking, logistics, and manufacturing). OOP principles (e.g., encapsulation, polymorphism) support **scalable, maintainable systems**. Java frameworks (Spring, Hibernate) are built on OOP and drive **innovation** in software infrastructure. (SDG 9). |
| 3 | **SDG 11:** Sustainable Cities and Communities | Java applications power **smart city solutions** (IoT, traffic systems). OOP enables creating modular systems: Traffic Light, Vehicle, Road Sensor. Promotes integration of **sustainable infrastructure monitoring** systems. (SDG 11). |

| 3rd Semester | Professional Core Course (PC)<br>LOGIC DESIGN AND COMPUTER ORGANIZATION | M25BCS303 |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Mathematics | Proficiency in algebra for Boolean expression simplification using K-map Techniques. |
| 2 | Fundamental Electronics Knowledge | Knowledge of number systems and basic digital logic gates (AND, OR, NOT, etc.).<br>Understanding of Boolean algebra and logic simplification techniques. |
| 3 | Programming Concepts of C programming | Basic understanding of functional and operational units of computer system for memory access. |
| 4 | Fundamental Electronics Knowledge | Basic understanding of electronic components, circuits, and their behavior.<br>Knowledge of binary number system and arithmetic. |

**2. Syllabus**

| LOGIC DESIGN AND COMPUTER ORGANIZATION<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS303** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **32:32:0:16:0 = 80Hrs** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**

1. Discuss the functionality of binary logic system
2. Illustrate the working of different combinational circuits
3. Analyze the working of different flip flops, registers and counters
4. Realize the basic concepts of computer system
5. Illustrate the working of I/O and memory units

**Module -1**

**Introduction to Digital Logic**: Binary Logic, Boolean algebra, Canonical and standard forms, digital logic gates, simplification of Boolean expressions using k-map techniques up to 4 variables (including don't-care condition), Product of sums simplification, NAND-NOR implementation, Tabulation method.

**Text1: 2.1 to 2.7, 3.1 to 3.3, 3.5, 3.6, 3.8, 3.9.** **8 Hours**

**Module -2**

**Combinational logic and its applications:** Introduction**,** Design procedure, adders, subtractors, Code conversions, Analysis procedure, multilevel NAND and NOR, Exclusive OR and Equivalence function, binary adder, decoder, encoder, multiplexer.

**Text1: 4.1 to 4.9, 5.1 to 5.2, 5.5, 5.6.** **8 Hours**

**Module -3**

**Sequential logic and its applications:** Introduction, flip flops, Triggering of flip flops, analysis of clocked sequential circuits, State reduction and assignment, flip flop excitation tables, Design of counters, analysis of clocked sequential circuits, ripple counters.

**Text1: 6.1 to 6.8, 7.4.** **8 Hours**

**Module -4**

**Fundamentals of computer structure and machine instructions:** Functional units,Basic Operational Units, Bus Structure, Performance, Memory location and addresses, Memory Operations, Instruction and Instruction sequencing, addressing modes, assembly language.

**Text 2: 1.2, 1.3, 1.4, 1.6(1.6.1, 1.6.2, 1.6.4, 1.6.7), 2.2 (except 2.2.4), 2.3, 2.4 (till 2.4.5), 2.5.** **8 Hours**

**Module -5**

**Organization of Input/output and memory unit:** Accessing I/O devices**,** Interrupts**,** Direct memory access, semiconductor RAM memories, Read only memories, speed, size and cost, cache memory and its mapping techniques, pipelining technique.

**Text2: 4.1, 4.2(till 4.2.3), 4.4, 5.2 (5.2.2,5.2.4), 5.3, 5.4, 5.5(except 5.5.4)** **8 Hours**

**Text Books:**

1.M. Morris Mano, "Digital Logic and Computer Design", Pearson, 2011

2. Carl Hamachar, ZvokoVranesic, SafwatZaky, "Computer Organization", 5thedition, Tata McGraw Hill, 2002

**Reference Books:**

1. Donald D Givonne, "Digital Principles and Design", Tata McGraw Hill, 2002.
2. Shuangbao Paul wang, Robert S. Ledley,"Computer Architecture and Security", 2013

### 3. Teaching-Learning Process Strategies

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 6 | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |

### 4. Assessment Details (both CIE and SEE)

**Continuous Internal Evaluation:**

**CIE Split up for Professional Course (PC)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B)* | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

***Note: Assessment plan:**

1. Two assignments will be conducted with 10 marks allotted for each assignment and average marks are considered for assessment.
2. Two Quizzes will be conducted via offline mode with four sets of 10 questions each and carries 10 marks. Average of Quiz marks is considered for assessment.
3. Activity for 5 marks is conducted on HDL programming for digital circuits**.**

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks.

### 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS303.1** | Apply the reduction techniques to solve the Boolean expressions. |
| **M25BCS303.2** | Apply reduction technique to design combinational circuits. |
| **M25BCS303.3** | Analyze the design of sequential circuits. |
| **M25BCS303.4** | Examine the performance measure using the concepts of I/O and memory units. |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS303.1** | 3 | - | - | - | - | - | - | - | - | - | - | - | - |
| **M25BCS303.2** | 3 | 3 | - | - | - | - | - | - | - | - | - | - | - |
| **M25BCS303.3** | - | 3 | 3 | - | - | - | - | - | - | - | - | - | - |
| **M25BCS303.4** | - | 3 | - | - | - | - | - | - | - | - | - | 3 | 3 |
| **M25BCS303** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **3** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4**: Quality Education | Teaching digital logic design develops problem-solving and critical thinking skills. Hands on learning using simulation and hardware kits for validating digital circuits. |
| 2 | **SDG7**: Affordable and clean energy | Efficient logic designs help reduce power consumption in devices. |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure. | Digital logic design is the backbone of computing hardware, supporting innovation in automation, robotics, AI, and IoT. |
| 4 | **SDG 11**: Sustainable cities and communities. | Enables development of smart city components such as traffic control systems, environmental monitoring, and public transport automation. Logic circuits in smart sensors used for waste management, weather monitoring or pollution control. |
| 5 | **SDG 12**: Responsible consumption and production | Promotes design of resource-efficient digital systems with minimal electronic waste. Encourages reusable modular logic designs to minimize environmental impact. |

| 3rd Semester | Professional Core Course (PC)<br>DATA STRUCTURE AND ITS APPLICATIONS | M25BCS304 |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic Programming Skills | Understanding of programming fundamentals, preferably in C or C++, including variables, control structures (loops, conditionals), functions, and arrays. |
| 2 | Mathematics | Basic knowledge of discrete mathematics, including concepts like sets, relations, functions, and logic, which help in understanding algorithms and data organization. |
| 3 | Problem-Solving Skills | Ability to analyze a problem, break it down into smaller parts, and devise step-by-step solutions or algorithms. |
| 4 | Understanding of Memory Concepts | Basic understanding of how memory works (variables stored in memory, pointers, dynamic memory allocation) to effectively work with data structures like linked lists, trees, etc. |

**2. Syllabus**

| DATA STRUCTURE AND ITS APPLICATIONS<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS304** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **32:32:0:16:20 = 100 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**

1. To explain fundamentals of data structures and their applications.
2. To illustrate representation of Different data structures such as Stack, Queues, Linked Lists, Trees and Graphs.
3. To Design and Develop Solutions to problems using Linear Data Structures
4. To discuss applications of Nonlinear Data Structures in problem solving.
5. To introduce advanced Data structure concepts such as Hashing, Trees and Optimal Binary Search Trees

**Module -1**

**Introduction To Data Structures**: Data Structures, Classifications (Primitive & Non-Primitive), Data structure Operations.

**ARRAYS And STRUCTURES:** Arrays, Dynamic Allocated Arrays, Structures and Unions, Polynomials, Sparse Matrices, representation of Multidimensional Arrays, Strings

**Text Book: Chapter-2: 2.1 to 2.6**

**Reference Book 1: 1.1 to 1.48 Hours**

**Module -2**

**STACKS:** Stacks, Stacks Using Dynamic Arrays, Evaluation and conversion of Expressions

**QUEUES**: Queues, Circular Queues, Using Dynamic Arrays, Multiple Stacks and queues.

**Text Book: Chapter-3: 3.1 to 3.4,3.6, 3.78 Hours**

**Module -3**

**LINKED LISTS:** Singly Linked, Lists and Chains, Representing Chains in C, Linked Stacks and Queues, Polynomials, Additional List Operations, Sparse Matrices, Doubly Linked List.

**Text Book: Chapter-4: 4.1 to 4.5,4.7,4.88 Hours**

**Module -4**

**TREES:** Introduction, Binary Trees, Binary Tree Traversals, Threaded Binary tree, Binary Search trees, Selection Trees, Forests.

**GRAPHS:** The Graph Abstract Data Types, Elementary Graph Operations.

**Text Book: Chapter-5: 5.1 to 5.3, 5.5, 5.7 to 5.9 Chapter-6: 6.1, 6.2** **8 Hours**

**Module -5**

**HASHING:** Introduction, Static Hashing, Dynamic Hashing

**EFFICIENT BINARY SEARCH TREES:** Optimal Binary Search Trees

**Tries:** Definition, Searching A Tries , Sampling Strategies, Insertion Into A Tries , Deletion From A Tries

**Text Book: Chapter 8: 8.1 to 8.3 Chapter 10: 10.1,10.9** **8 Hours**

**TEXTBOOKS:**

1.Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed, Fundamentals of Data Structures in C, 2nd Ed, Universities Press, 2014

2. Seymour Lipschutz, Data Structures Schaum's Outlines, Revised 1st Ed, McGraw Hill, 2014.

**REFERENCE BOOKS:**

1. Gilberg & Forouzan, Data Structures: A Pseudocode approach with C, 2nd Ed, Cengage Learning,2014.

2. Reema Thareja, Data Structures using C, 3rd Ed, Oxford press, 2012.

**VIDEO LINKS:**

1. http://www.nptelvideos.in/2012/11/data-structures-and-algorithms.html
2. https://mitlibraryblog.wordpress.com/mitm-videos/

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of various data structure concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 6 | Programming Assignments | Assign programming tasks to reinforce practical skills associated with competencies. |

**4. Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Professional Course (PC)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B)* | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work Marks

*** NOTE: Assessment Tools:**

**1.** Programming Assignments

Purpose: To assess coding proficiency in implementing data structures.

Tools Used:

- C in IDEs like Code=Blocks, Eclipse, or VS Code.
- Online platforms like JDoodle, OnlineGDB, CodeChef, or Replit.

2.Quiz (Online/Offline)

Purpose: For formative assessment and continuous evaluation.

Platforms: Google Forms, Kahoot, Moodle, Blackboard, etc.

Content:

- MCQs on time complexity.
- True/False on data structure properties.
- Matching algorithms to their suitable use-cases.

**Self-Learning (SL): If applicable, t**he teaching faculty shall motivate the students to take up online courses from any recognized platforms. There shall not be any assessment of the Self-Learning component. The faculty must collect the certificate from the students who have successfully completed the self-learning relevant to the course.

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20 marks. The medium of the

question paper shall be English unless otherwise it is mentioned.

2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS304.1** | **Apply** appropriate linear data structures such as arrays, stacks, queues, and structures to solve computational and real-life problems. |
| **M25BCS304.2** | **Apply** linked lists and their variants in implementing dynamic data storage and algorithmic solutions like polynomial operations. |
| **M25BCS304.3** | **Analyze**tree and graph data structures for organizing hierarchical and networked data, including traversal and manipulation techniques. |
| **M25BCS304.4** | **Evaluate** and use advanced data structures like hashing, optimal binary search trees, and tries for efficient data access and retrieval |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS304.1** | 3 | 2 | 3 | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS304.2** | 3 | 2 | 3 | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS304.3** | 3 | 3 | 2 | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS304.4** | 3 | 3 | 3 | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS304** | **3** | **2.5** | **2.75** | - | - | - | - | - | - | - | - | **3** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4**: Quality Education | The course builds critical computational and analytical thinking skills, empowering students with lifelong learning tools in computer science. Promotes technical education and employability |
| 2 | **SDG 8**: Decent Work and Economic Growth | By equipping students with key programming and problem-solving skills, the course enhances career opportunities in IT and software sectors, contributing to economic growth. |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure | Data structures form the backbone of modern technology and software infrastructure. The course fosters innovation and prepares students for building efficient, scalable digital systems |

| 3rd Semester | Engineering Science Course (ES)<br>UNIX AND SHELL PROGRAMMING | M25BCS305A |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic Computer Skills | Familiarity with using a computer, navigating the file system, managing files and directories, and using the command line interface (CLI) is essential. |
| 2 | Understanding of Operating Systems | Basic knowledge of how operating systems work, including processes, memory management, file systems, and user permissions. |
| 3 | Familiarity with Command Line Interface (CLI) | Understanding of how to navigate directories, list files, create directories, copy/move files, and execute commands using the command line interface. |
| 4 | Programming Fundamentals | Basic understanding of programming concepts like variables, data types, loops, conditionals, functions, and control structures. This will help in understanding shell scripting. |
| 5 | Text Editing Skills | Proficiency in using a text editor, as shell scripts are essentially text files containing commands. |

**2. Syllabus**

| UNIX AND SHELL PROGRAMMING<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS305A** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **32:32:0:16:20 = 100 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**

The main objectives of this course are to:

- Understand the features, architecture of UNIX and its commands.
- Discuss different UNIX files, attributes and permissions.
- Discuss filter programs and regular expressions.
- Familiarize with advanced filters

**Module -1**

**UNIX Architecture and Command Usage:** UNIX Architecture, Features of UNIX, Internal and External Commands. General-Purpose Utilities: cal, date, echo, printf, bc, passwd, who, uname, tty, stty. **The File System**: The Parent-Child Relationship, the HOME variable, pwd, cd, mkdir, rmdir, Absolute Pathnames,Relative Pathnames, ls: Listing Directory Contents.

**(Text book: 1 Chapters: 2.1, 2.2, 2.5, 3.1 to 3.5, 3.9 to 3.13, 4.1 to 4.11)8 Hours**

**Module -2**

**Handling Ordinary Files**: cat, cp, rm, mv, more, file, wc, od, cmp, comm, diff, Basic File Attributes: ls –l, the –d option, file ownership, file permissions, chmod, directory permissions, changing file ownership. More **File Attributes**: File Systems and Inodes, Hard Links, Symbolic Links and ln, umask Modification and Access Times.

**(Text book: 1 Chapters: 5.1 to 5.5, 5.7 to 5.12, 6.1 to 6.7, 11.1 to 11.6)8 Hours**

**Module -3**

**Simple Filters**: The sample database, paginating files, head, tail, cut, paste, sort, uniq, tr. **Filters using RegularExpression**: grep: Searching for a pattern, Basic Regular Expression (BRE), egrep: Extended Regular Expression. **(Text book: 1 Chapters: 12.1, 12.9 to 12.9, 13.1 to 13.3)8 Hours**

**Module -4**

**Essential Shell Programming Part I**: Shell Scripts, read, Using command line arguments, exit and exit status of command, the logical operators && and || - conditional execution, the if conditional, using test and [ ] to evaluate expressions, the case conditional, expr, $0, while, for, set and shift, trap: Interrupting a program

**(Text book: 1 Chapters: 14.1 to 14.13)8 Hours**

**Module -5**

**awk: An Advanced Filter**: Simple awk filtering, Splitting a Line into Fields, printf: Formatting Output, Variables and Expressions, Comparison Operators, Number Processing, Built-in Variable, Arrays,

| |
|---|
| Functions, Control Flow, Looping with for and while<br>**(Text book: 1 Chapters: 18.1 to 18.15).8 Hours** |
| **TEXTBOOKS:**<br>1. Sumitabha Das, "UNIX – Concepts and Applications", 4th Edition, McGraw Hill, 2017.<br>2. Behrouz A Forouzan and Richard F Gilberg, "UNIX and SHELL Programming", India. Edition, Cengage Learning, Third Reprint 2008<br>**REFERENCE BOOKS:**<br>1. Kenneth Rosen et al, "UNIX – The Complete Reference", 2nd Edition, Tata McGraw Hill Fourth Reprint 2008<br>2. Sumitabha Das, "Your UNIX: The Ultimate Guide", McGraw Hill, 2001.<br>3. M G Venkateshmurthy, "Introduction to UNIX and Shell Programming", Pearson Edition. |

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Live Demonstration | Develop and run Shell scripts in the classroom. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |
| 5 | Problem-Based Learning (PBL) | Implement PBL to enhance analytical skills and practical application of competencies |
| 6 | Multiple Representations | Introduce topics in various representations to reinforce competencies |
| 7 | Programming Assignments | Assign programming tasks to improve the practical skills. |

**4. Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Engineering Science Course (ESC)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment - Tests (A) | 3 | 50% | 25 |
| 2 | Term Work - TW (B) * | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

*** Assessment / Case Studies / Practices**

**1.** Assignment and Quiz

2. Problem Based

3. Group presentation (4 members in a team)

**Self-Learning (SL): If applicable, t**he teaching faculty shall motivate the students to take up online courses from any recognized platforms. There shall not be any assessment of the Self-Learning component. The faculty must collect the certificate from the students who have successfully completed the self-learning relevant to the course.

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20 marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS305A.1** | **Understand** the architecture and features of UNIX operating system and its file system**.** |

| | |
|---|---|
| **M25BCS305A.2** | **Apply** UNIX commands for file processing, process management, and system utilities. |
| **M25BCS305A.3** | **Develop** shell scripts using various control structures, loops, and command-line arguments. |
| **M25BCS305A.4** | **Demonstrate** proficiency in handling filters, pipes, regular expressions, and text processing tools like awk, sed, and grep |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS305A.1** | 3 | - | - | - | 3 | - | - | - | - | - | 2 | - | - |
| **M25BCS305A.2** | 3 | - | - | - | 3 | - | - | - | - | - | 2 | 2 | - |
| **M25BCS305A.3** | | 3 | 3 | - | 3 | - | - | - | - | - | 2 | - | - |
| **M25BCS305A.4** | | 3 | 2 | 3 | 3 | - | - | - | - | - | 2 | - | 2 |
| **M25BCS305A** | **3** | **3** | **2.5** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **2** | **2** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4** – Quality Education | Increase the number of youth and adults with relevant skills, including technical and vocational skills, for employment and entrepreneurship. |
| 2 | **SDG 8** – Decent Work and Economic Growth | UNIX and shell scripting are widely used in system administration, DevOps, and automation jobs. |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure, | Encourages innovation through scripting for process automation, system monitoring, and efficient computing. |

| 3rd Semester | Engineering Science Course (ES)<br>DOT NET PROGRAMMING | M25BCS305B |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic C# Programming | Ability to write, debug, and structure simple C# programs using variables, methods, and control flow. |
| 2 | Object-Oriented Concepts | Understanding and implementing OOP principles such as classes, inheritance, interfaces, and encapsulation. |
| 3 | Data Structures & Collections | Proficient in using arrays, generics, collections, and managing memory with garbage collection. |
| 4 | Event Handling & LINQ | Capable of decoupling application logic using events and querying in-memory data using LINQ. |

**2. Syllabus**

| DOT NET PROGRAMMING<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS305B** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L: T: P:TW:SL) | **32:32:0:16:20 = 100 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**

1. Inspect Visual Studio programming environment and toolset designed to build applications for Microsoft Windows
2. Understand Object Oriented Programming concepts in C# programming language.
3. Interpret Interfaces and define custom interfaces for application.
4. Build custom collections and generics in C#
5. Construct events and query data using query expressions

**Module -1**

**Introducing Microsoft Visual C# and Microsoft Visual Studio 2015:** Welcome to C#, Working with variables, operators and expressions, Writing methods and applying scope, Using decision statements, Using compound assignment and iteration statements, Managing errors and exceptions.
**T1:Chapter 1-Chapter 68 Hours**

**Module -2**

**Understanding the C# object model:** Creating and Managing classes and objects, Understanding values and references, Creating value types with enumerations and structures, Using arrays.
**T1:Chapter 7-Chapter 108 Hours**

**Module -3**

Understanding parameter arrays, Working with inheritance, Creating interfaces and defining abstract classes, Using garbage collection and resource management.
**T1:Chapter 11-Chapter148 Hours**

**Module -4**

**Defining Extensible Types with C#:** Implementing properties to access fields, Using indexers, Introducing generics, Using collections.
**T1: Chapter15- Chapter188 Hours**

**Module -5**

Enumerating Collections, Decoupling application logic and handling events, Querying in- memory data by using query expressions, Operator overloading.
**T1:Chapter 19- Chapter 228 Hours**

**TEXTBOOKS:**

1. John Sharp, "Microsoft Visual C# Step by Step", 8th Edition, PHI Learning Pvt. Ltd. 2016.
2. Andrew Troelsen, "Prof C# 5.0 and the .NET 4.5 Framework", 6th Edition, Apress and Dreamtech Press, 2012.

**REFERENCE BOOKS:**

1. Christian Nagel, "C# 6 and .NET Core 1.0", 1st Edition, Wiley India Pvt Ltd, 2016.
Andrew Stellman and Jennifer Greene, "Head First C#", 3rd Edition, O"ReillyPublications,

2013.
2. Mark Michaelis, "Essential C# 6.0", 5th Edition, Pearson Education India, 2016.

**VIDEO LINKS:**

1. https://www.youtube.com/watch?v=GhQdlIFylQ8
2. https://www.youtube.com/watch?v=0QUgvfuKvWU

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of Dot().Net Concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 6 | Laboratory Learning | Utilize the facilities available in the laboratories to understand the behavior of the materials by performing few experiments. |

**4. Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Engineering Science Course (ES)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B) | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

*** Assessment / Case Studies / Practices**

**1.** Assignment and Quiz
2. Problem based approach
3. Group Presentation

**Self-Learning (SL): If applicable, t**he teaching faculty shall motivate the students to take up online courses from any recognized platforms. There shall not be any assessment of the Self-Learning component. The faculty must collect the certificate from the students who have successfully completed the self-learning relevant to the course.

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20 marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 questions from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS305B.1** | **Understand and Apply** applications on Visual Studio .NET platform by understanding the syntax and semantics of C#. |
| **M25BCS305B.2** | **Apply** Object Oriented Programming concepts in C# programming language. |
| **M25BCS305B.3** | **Analyze** custom interfaces for applications and leverage the available built-in interfaces in building complex applications. |

| | |
|---|---|
| **M25BCS305B.4** | **Design** the use of generics and collections in C#. |
| **M25BCS305B.5** | **Implement** queries to query in-memory data and define own operator behaviour. |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS305B.1** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS305B.2** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS305B.3** | - | 3 | - | - | - | - | - | - | - | - | - | 3 | 3 |
| **M25BCS305B.4** | - | - | 3 | - | - | - | - | - | - | - | - | - | 3 |
| **M25BCS305B.5** | - | - | - | 3 | - | - | - | - | - | - | - | - | 3 |
| **M25BCS305B** | **3** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **3** |

**6. Mapping to Sustainable Development Goals (SDG)**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4** Quality Education | Promotes skill development through technical programming education. |
| 2 | **SDG 8** Decent Work and Economic Growth | Equips students with industry-relevant programming knowledge for employment. |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure, | Supports innovation by teaching modern, scalable coding practices using C#. |

| 3rd Semester | Engineering Science Course (ES)<br>OOPS WITH C++ | M25BCS305C |
|---|---|---|

**1. Prerequisites**

| S/L | Proficiency | Prerequisites |
|---|---|---|
| **1.** | Basic Computer Skills | Familiarity of different Operating Systems and the basic knowledge of command line usage are very needful. |
| **2.** | Knowledge of Integrated Development Environment | Requires the basic skills to use various tools like text editor, compiler, linker and C++ IDE. |
| **3.** | Problem Solving Skills | Knowledge of the Algorithmic thinking and Logical thinking needed. |
| **4.** | Mathematics | Proficiency in Mathematics required to find the roots of quadratic equation, Trigonometric Functions etc. |
| **5.** | Basics of C Programming | Fundamental understanding of C is essential for object-oriented programming. This includes syntax, data types, variables, control structures, functions, and pointers |

**2. Syllabus**

| **OOPS WITH C++<br>SEMESTER – III** | | | |
|---|---|---|---|
| Course Code | **M25BCS305C** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem (L:T: P:TW:SL) | **32:32:0:16:20 = 100 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**
1. To learn primitive constructs C++ programming language.
2. To understand Object Oriented Programming Features of C++.
3. To gain knowledge on: Exceptions and C++ I/O System Basics.

**Module -1**

**Introduction to object Oriented Programming**: OOP Paradigm, Basic concepts of OOP, Beginning with C++, Applications of C++, A simple C++ programs, Structure of C++ Program.
**Tokens:** Character sets and Symbols, Keywords, C++ Identifiers, Variables and Constants, Dynamic Initialization of variables, Reference variables, Basic data types in C++, User defined data types, Storage classes, Operators, Type cast Operators.
**Chapter 1,2,3** **8 Hours**

**Module -2**

**Decision and Control Structures:** if statement, if-else statement, switches statement, Loop: while, do while, for, Jump Statements: break, return, go to.
**Classes and Objects:** Classes in C, class declaration, declaring objects, Define member functions, call by reference, return by reference, inline functions, default arguments, Function Overloading.
**Chapter 4,5** **8 Hours**

**Module -3**

**Constructor and Destructors:** Constructors, Parameterized constructors, Multiple Constructors in a class, Constructors with default arguments, Dynamic initialization of Objects, Const object, Destructors.
**Operator Overloading:** Introduction, Defining operator overloading, Overloading unary and binary operators, Type Conversions.
**Chapter 6,7** **8 Hours**

**Module -4**

**Inheritance:** Defining Derived classes, Types of Inheritance- Single inheritance, Multilevel inheritance, Multiple inheritance, Hierarchical inheritance, Hybrid Inheritance, Abstract classes, constructors in derived class, Member classes. **Polymorphism:** Introduction, Virtual functions, virtual constructor and destructors.
**Chapter 8,9** **8 Hours**

**Module -5**

**Exception Handling**: Basic of Exception Handling, Exception Handling Mechanism, Throwing Mechanism, Catching Mechanism, Rethrowing an Exception, Exception in Operator overloaded functions.
**The C++ I/O System Basics:** C++ Streams, The C++ Classes, Formatted I/O File I/O: and File Classes, Opening and Closing a File, Reading and Writing Text Files, Detecting EOF.
**Chapter 11,13** **8 Hours**

**Text Books:**

1. Balagurusamy E, Object Oriented Programming with C++, Tata McGraw Hill Education Pvt.Ltd. Sixth Edition 2016.
2. Herbert schildt, The Complete Reference C++, 4th edition, TMH, 2005.

**Reference Books:**

1. Herbert schildt, The Complete Reference C++, 4th edition, TMH, 2005
2. D.S Guru, Object- Oriented Programming with C++.

**VIDEO LINKS:**

1. https://onlinecourses.nptel.ac.in/noc22_ce54/preview
2. https://mitlibraryblog.wordpress.com/mitm-videos/

### 3. Teaching-Learning Process Strategies

| S/L | TLP Strategies: | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |
| 5 | Problem-Based Learning (PBL) | Implement PBL to enhance analytical skills and practical application of competencies |
| 6 | Multiple Representations | Introduce topics in various representations to reinforce competencies |
| 7 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 8 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 9 | Programming Assignments | Assign programming tasks to reinforce practical skills associated with competencies. |

### 4. Assessment Details (both CIE and SEE)

**Continuous Internal Evaluation:**

**CIE Split up for Engineering Science Course (ESC)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B)* | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks =(A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

*** Assessment / Case Studies / Practices**

**1.** Assignment and Quiz
2. Problem Based
3. Group presentation (4 members in a team)

**Self-Learning (SL): If applicable, t**he teaching faculty shall motivate the students to take up online courses from any recognized platforms. There shall not be any assessment of the Self-Learning component. The faculty must collect the certificate from the students who have successfully completed the self-learning relevant to the course.

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 question from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

## 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS305C.1** | **Understand** and **Apply** the basic programming constructs. |
| **M25BCS305C.2** | **Apply** the structure of classes and methods in C++ programming environment. |
| **M25BCS305C.3** | **Analyse** the different programming constructs of C++ and its effectiveness in improving the efficiency of C++ programs. |
| **M25BCS305C.4** | **Implement** appropriate C++ programming constructs to solve real-world problem sample scenarios. |
| **M25BCS305C.5** | Use the concepts of input output streams for file operations |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS305C.1** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS305C.2** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS305C.3** | - | 3 | - | - | - | - | - | - | - | - | - | - | 3 |
| **M25BCS305C.4** | - | - | 3 | - | - | - | - | - | - | - | - | - | 3 |
| **M25BCS305C.5** | - | - | - | 3 | - | - | - | - | - | - | - | - | 3 |
| **M25BCS305C** | **3** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **3** |

## 6. Mapping to Sustainable Development Goals (SDG):

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 3:** Good Health and Well-being | Covering aspects of health, hygiene, and well-being relevant to the syllabus |
| 2 | **SDG 4:** Quality Education | Emphasizing inclusive and equitable quality education and promoting lifelong learning opportunities for all |
| 3 | **SDG 13:** Climate Action | Addressing environmental awareness, conservation, and sustainable practices included in the curriculum. |

| 3rd Semester | Engineering Science Course (ES)<br>APPLIED MATHEMATICS FOR DATA SCIENCE | M25BMATD305D |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic concepts of Statistic and Curve fitting | Understanding the relationship between one or more predictors and a response variable with the goal of defining a "best fit" model of the relationship Familiarity with fundamental knowledge of algebra course |
| 2 | Calculus | Knowledge of calculus, specifically integration and differentiation and an understanding of complex numbers |
| 3 | Basic Concepts of Linear Algebra | Strong knowledge of calculus, linear algebra, complex numbers and trigonometric function |
| 4 | Basic Concept of Permutation and Combination | Knowledge of basics set theory, inclusion and exclusion principle (knowing different ways of counting) and calculus(knowing derivatives and integrals ) |

**2. Syllabus**

| APPLIED MATHEMATICS FOR DATA SCIENCE<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BMATD305D** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem (L:T: P:TW:SL) | **32:32:0:16:20 = 100 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **03** | Exam Hours | **03** |

**Course Objectives:**

1. To have insight in to Statistical methods, correlation and regression analysis, fitting of curves
2. To Provide the principles of statistical inferences and the basics of hypothesis testing with emphasis on some commonly encountered hypotheses.
3. To Determine whether an input has a statistically significant effect on the system's response through ANOVA testing.

**Module -1** **8 Hours**

**Statistical Methods and Curve Fitting:** Curve fitting by the method of least squares, fitting the curve of the forms $y = ax + b, y = ax^b$ and $y = ax^2 + bx + c$

Correlation and regression- Karl Pearson's coefficient of correlation and rank correlation, problems. Regression analysis, lines of regression, problems

**Module -2** **8Hours**

**Linear Algebra:** Eigen values and Eigen Vectors, Diagonalization, Linear Transformation, Inverse Linear Transformation Orthogonal Transformations, Quadratic to canonical form (2nd& 3rd order)

**Module -3** **8Hours**

**Probability Distribution:** Review of basic probability theory. Random variables (discrete and continuous), probability mass and density functions. Mathematical expectation, mean and variance. Binomial, Poisson Exponential and Normal Distributions, (Statements only), Problems

Joint probability distribution: Joint Probability distribution for two discrete random variables, Expectation, covariance and correlation

**Module -4** **8Hours**

**Statistical Inference:** Sampling variables, central limit theorem and confidences limit for unknown mean. Test of

Significance for means of two small samples, Students't' distribution, Chi-square distribution as a test of goodness of fit. F-Distribution..

**Module -5** **8Hours**

**Design of Experiments & ANOVA:** Principles of experimentation in design, Analysis of completely randomized design, randomized block design. The ANOVA Technique, Basic Principle of ANOVA, One-way ANOVA, Two-way ANOVA, Latin-square Design

**TEXTBOOKS:**

1. JB.S.Grewal:"Higher Engineering Mathematics",Khanna publishers,44th Ed.2018
2. T. Veerarajan: Probability, Statistics and Random Processes with Queuing theory and Queuing Networks, McGraw Hill Education, Fourth Edition

**REFERENCE BOOKS:**

1. 1. S. M. Ross, "A first course in Probability", Prentice Hall.

**3.Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of the concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |

4. **Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Engineering Science Course(ES)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B)* | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

*** Assessment/Case Studies/Practices**

1. Assignment
2. Case Study Analysis
3. Practice Problems

**Self-Learning (SL): If applicable, t**he teaching faculty shall motivate the students to take up online courses from any recognized platforms. There shall not be any assessment of the Self-Learning component. The faculty must collect the certificate from the students who have successfully completed the self-learning relevant to the course.

**Semester End Examination:**

1. Question paper pattern will be ten questions. Each question is set for 20marks. The medium of the question paper shall be English unless otherwise it is mentioned.
2. There shall be 2 question from each module, each of the two questions under a module (with a maximum of 3 sub questions), may have mix of topics under that module if necessary.
3. The students have to answer 5 full questions selecting one full question from each module.
4. Marks scored will be proportionally scaled down to 50 marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BMATD305D.1** | Apply the concepts of Statistics, Probability theory & ANOVA to solve CSE problems |
| **M25BMATD305D.2** | Analyze the concept of Statistics , Probability theory and ANOVA |
| **M25BMATD305D.3** | Validate Computer Science & Engineering Applications in Statistics Probability theory and ANOVA |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BMATD305D.1** | 3 | | - | - | - | - | - | - | - | - | - | 2 | 3 |
| **M25BMATD305D.2** | - | 3 | | - | - | - | - | - | - | - | - | 2 | 3 |
| **M25BMATD305D.3** | 3 | | 3 | - | - | - | - | - | - | - | - | 2 | 3 |
| **M25BMATD305D** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **2** | **3** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 1**: No Poverty | Statistical models can estimate the likelihood of individuals or communities falling below the poverty line. These models help policymakers target resources more effectively and identify at-risk populations. |
| 2 | **SDG 2** – Zero Hunger | Supply Chain Optimization: In food security, linear transformations can help optimize supply chain networks, ensuring that food resources are distributed effectively to regions in need. By modelling the flow of food from producers to consumers, matrix transformations can be used to optimize logistics and reduce waste. |
| 3 | **SDG 11 –** Sustainable Cities and Communities | **ANOVA for Urban Sustainability Projects**: Different urban sustainability initiatives (e.g., green buildings, recycling programs, transportation changes) can be compared using ANOVA to assess their effectiveness. |

| 3rd Semester | Professional Core Course Laboratory (PCL)<br>DATA STRUCTURES LABORATORY | M25BCSL306 |
|---|---|---|

**1. Prerequisites**

| S/L | Proficiency | Prerequisites |
|---|---|---|
| 1 | Fundamental Programming Skills: | • Proficiency in at least one programming language: Often, languages like C, C++, Java, or Python are preferred.<br>• Understanding of basic programming constructs: Loops, conditionals, functions, and recursion. |
| 2 | Basic Algorithm Understanding: | • Sorting and searching algorithms. |
| 3 | Specific Topics | • Understanding of primitive data types and structures like arrays, linked lists, stacks, queues, and hash tables.<br>• Basic understanding of memory allocation and pointers (especially for languages like C and C++).<br>• Ability to implement recursive and iterative solutions. |
| 4 | Debugging and Compilation Skills | • Skilled at identifying errors and debugging code in an IDE or terminal |

**2. Syllabus**

| DATA STRUCTURES LABORATORY<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCSL306** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem (L:T: P:TW:SL) | **0:0:32:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course objectives:** This laboratory course enables students to get practical experience in design, develop, implement, analyze and evaluation/testing of

1. Dynamic memory management
2. Linear data structures and their applications such as stacks, queues and lists
3. Non-Linear data structures and their applications such as trees and graphs

| Sl. No | Experiments |
|---|---|
| | **PRACTICAL COMPONENT**<br>Implement all the programs in C Programming Language |
| 1 | Write a program in C using array to:<br>• Accept marks for 'n' students in 'm' subjects.<br>• Display individual total and Average marks.<br>• Find student with highest and lowest total marks.<br>• Display subject-wise highest and average marks. |
| 2 | Write a C program to implement the Knuth-Morris-Pratt (KMP) string matching algorithm. The program should include:<br>• A function to preprocess the pattern and create the match table.<br>• A function to search for all occurrences of the pattern in a given text using the KMP algorithm. |
| 3 | Create a menu-driven program in C to perform various operations on a stack of integers using an array-based implementation. The stack should have a maximum size defined as MAX. Perform the following operations.<br>• Push an Element on to Stack.<br>• Pop an Element from Stack.<br>• Demonstrate how stack can be used to check Palindrome.<br>• Demonstrate Overflow and Underflow situations on Stack.<br>• Display the status of Stack.<br>• Exit |
| 4 | Implement a Program in C for converting an Infix Expression to Postfix Expression. Program should support for both parenthesized and free parenthesized expressions with the operators: +, -, *, /, % (Modulus), ^ (Power) and alphanumeric operands. Convert this Infix Expression (a+ (b-c) *d) to Postfix Expression. |
| 5 | Write a C program for a Bank system using queue that allows:<br>• Customers to join the queue(enqueuer)<br>• Customers to be served(dequeuer) |

| | |
|---|---|
| | • Display of the queue at any time<br>• Count of customers waiting in line. |
| 6 | Create and implement a C program that provides a menu-driven interface for managing a circular queue of characters using an array-based approach. The circular queue should have a predefined maximum size, defined as MAX. Perform the following operations.<br>• Enqueue: Add an element to the queue.<br>• Dequeue: Remove an element from the queue.<br>• Display: Show all elements currently in the queue. |
| 7 | Create a menu-driven program in C for dynamic list of student records using Singly Linked List operations<br>• Add new student.<br>• Delete a student by USN.<br>• Display all students.<br>• Count total number of students. |
| 8 | Develop a menu-driven program in C for performing various operations on a Binary Search Tree (BST) of integers. The operations include:<br>• Creating a BST with the following integers: 6, 9, 5, 2, 8, 15, 24, 14, 7, 8, 5, 2<br>• Traversing the BST in In-order, Pre-order, and Post-order<br>• Searching the BST for a given element (KEY) and displaying an appropriate message<br>• Displaying the height of the binary tree<br>• Exiting the program |
| 9 | Write a C program to covert skewed binary tree into Optimal Binary Search Tree (OBST) using Dynamic Programming, given the key values and their search frequencies. |
| 10 | Write a program in C to perform the following tasks on a graph representing N cities:<br>• Create the graph using an adjacency matrix.<br>• Print all the nodes reachable from a given starting node in a directed graph using either Depth First Search (DFS) or Breadth First Search (BFS) methods. |
| 11 | Write a C program to implement trie(Prefix Tree) for the following operations:<br>• Insert words<br>• Search for a word<br>• Display all word in dictionary order. |
| 12 | Design a hash table for storing and retrieving students ID's. Implement insertion, search and display operation using Linear Probing for handling collisions. |

**Practice Questions to be implemented using C programming (**Not be considered for exam**)**

1. Implement a dynamic array class that supports operations like adding an element, removing an element, and resizing the array when it's full.
2. Implement the binary search algorithm on a sorted array of integers.
3. Write a program to multiply two matrices and display the result.
4. Implement a stack using a linked list and perform push, pop, and display operations.
5. Implementation of circular queue to manage tasks in a Round-Robin fashion.
6. Implement a function to add two polynomials and return the result.
7. Implement a graph using an adjacency list and an adjacency matrix.
8. Implement a graph traversal algorithm (BFS and DFS) for campus navigation and connectivity analysis.
9. Given a sorted array of book ID's, implement a binary search to determine whether a given book ID exists and if so, return its index.

**3. Teaching-Learning Process Strategies**

| S/L | TLP Strategies: | Description |
|---|---|---|
| 1 | Interactive Lectures | Use interactive lectures to introduce new concepts. Incorporate questions and discussions to engage students. |
| 2 | Coding Sessions | Demonstrate the implementation of data structures and algorithms live, showing step-by-step coding and debugging. |
| 3 | Lab Exercises | Design lab exercises that require students to implement and manipulate data structures. |
| 4 | Coding Assignments | Assign regular coding tasks that reinforce lecture material and provide practical experience. |
| 5 | Group Projects | Encourage students to work in groups for larger projects, fostering teamwork and collaborative problem-solving. |

| | | |
|---|---|---|
| 6 | Code Documentation: | Practice writing clear and comprehensive documentation for all coding assignments and projects. |

**4. Assessment Details (both CIE and SEE)**

**Test Marks distribution for the Practical Course**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Write-up, Conduction, Result and Procedure/Algorithm/Flowchart | 60% | 60 |
| 2 | Viva-Voce | 40% | 40 |
| **Total** | | **100%** | **100** |

**Final CIE in Practical Course including AEC:**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Scaled Down marks of conduction, record/journal | 60% of the maximum | 30 |
| 2 | Scaled-down marks of the test | 40% of the maximum | 20 |
| **Total** | | **100%** | **50** |

**SEE for the Practical Course:**

1. SEE marks for a practical course shall be 50 marks.
2. The practical course is evaluated for 100 marks, and the scored marks shall be scaled down to 50 marks.
3. A change of experiment/program is allowed only once, and 20% of the marks allotted to the procedure/write-up part will be zero.
4. The duration of SEE shall be 3 hours.

**Marks distribution for the Practical Course**

| SL. No. | Description | % Marks | Marks |
|---|---|---|---|
| 1 | Write-up, Procedure | 20% | 20 |
| 2 | Conduction and result | 60% | 60 |
| 3 | Viva-Voce | 20% | 20 |
| **Total** | | **100%** | **100** |

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCSL306.1** | Analyze and compare various linear and non-linear data structures |
| **M25BCSL306.2** | Code, debug and demonstrate the working nature of different types of data structures and their applications |
| **M25BCSL306.3** | Implement, analyse, and evaluate the searching and sorting algorithms |
| **M25BCSL306.4** | Choose the appropriate data structure for solving real world problems |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCSL306.1** | 3 | 2 | - | - | - | - | - | - | - | - | - | 3 | 2 |
| **M25BCSL306.2** | 3 | 3 | 2 | - | 2 | - | - | - | - | - | - | 3 | 3 |
| **M25BCSL306.3** | 3 | 2 | 3 | - | 2 | - | - | - | - | - | - | 3 | 3 |
| **M25BCSL306.4** | 3 | 3 | 3 | 2 | 2 | - | - | - | - | - | - | 3 | 3 |
| **M25BCSL306** | **3** | **2.5** | **2.66** | **2** | **2** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **2.75** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| **1** | **SDG-4:** Quality Education | Enables students with computational thinking and data problem-solving skills. |
| **2** | **SDG 8**: Decent Work and Economic Growth | Improves job-readiness with practical and analytical skills in data handling and performance. |
| 3 | **SDG 9**: Industry, Innovation, and Infrastructure | Trains students in optimizing data operations relevant for software industries. |

| | | |
|---|---|---|
| **3rd Semester** | **Ability Enhancement Course (AE)<br>DATA VISUALIZATION WITH PYTHON** | **M25BCS307A** |

**1. Prerequisites**

| S/L | Proficiency | Prerequisites |
|---|---|---|
| 1 | Mathematics | Basic Mathematics is necessary |
| 2 | Programming | Libraries: Familiarize yourself with libraries like Pandas (for data manipulation), NumPy (for numerical computations), C Programming |
| 3 | Data Manipulation | Data Visualization: Presenting data effectively using plots, charts, and graphs. |
| 4 | Domain Knowledge | Depending on your interests, having some knowledge in the field you want to apply Data Visualization to (e.g., finance, healthcare, marketing) can be beneficial. |
| 5 | Critical Thinking and Problem-Solving | Data Visualization often involves complex problems that require critical thinking and creative solutions. Practice problem-solving skills and logical reasoning. |

**2. Syllabus**

| **DATA VISUALIZATION WITH PYTHON<br>SEMESTER – III** | | | |
|---|---|---|---|
| Course Code | **M25BCS307A** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L: T: P:TW:SL) | **0: 0:32:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course objectives:** This course will enable students to:

1. Demonstrate the use of IDLE or PyCharm IDE to create Python Applications
2. Using Python programming language to develop programs for solving real-world problems
3. Implementation of Mat plot lib for drawing different Plots
4. Demonstrate working with Seaborn, Bokeh.
5. Working with Plotly for 3D, Time Series and Maps.

| Sl. No. | Experiments<br>List of problems for which student should develop program and execute in the Laboratory |
|---|---|
| 1 | a. Monthly sales data for five different products in a store is available. Write a Python program to visualize this data using a bar plot to compare the sales of each product.<br>b. Data on the ages and corresponding incomes of a group of individuals is given. Write a Python program to create a scatter plot to analyze the relationship between age and income. |
| 2 | a. A dataset contains the test scores of 50 students. Write a Python program to create a histogram to show the distribution of these test scores.<br>b. Market share data of five smartphone brands is given. Write a Python program to visualize this data using a pie chart to illustrate the proportion of each brand's market share. |
| 3 | a. A dataset contains the test scores of 50 students. Write a Python program to create a histogram to show the distribution of these test scores.<br>b. Market share data of five smartphone brands is given. Write a Python program to visualize this data using a pie chart to illustrate the proportion of each brand's market share. |
| 4 | A dataset contains customer purchase behavior with multiple variables. Write a Python program to create and customize Seaborn plots using aesthetic functions to enhance the visualization's appearance and clarity. |
| 5 | a. Time-series data of website traffic over a week is available. Write a Python program to create a Bokeh line graph with annotations and legends to highlight key events.<br>b. Various datasets containing sales, profits, and customer counts are provided. Write a Python program to plot different types of visualizations using Bokeh to analyze the data. |
| 6 | A dataset contains three-dimensional coordinates representing the positions of geological samples. Write a Python program to visualize these points using 3D plots with Plotly libraries. |
| 7 | a. Historical stock price data over several months is given. Write a Python program to visualize this data as a time series using Plotly libraries.<br>b. Geographic data showing locations of retail stores across a country is available. Write a Python program to create an interactive map visualization using Plotly libraries. |
| 8 | Given immigration data to Canada from 1980 to 2013:<br>• Create an area plot for the top 6 immigrant countries from 1990 to 2013.<br>• Create a year-wise bar chart showing immigrants from India to Canada during 1980 to |

| | |
|---|---|
| | 2013.<br>• Create a boxplot for immigrants from India, Philippines, and China.<br>• Visualize the total number of immigrants from India and France using an area chart and a pie chart.<br>• Create a scatter histogram for immigrants from Fiji and Singapore in the year 2013.<br>https://www.kaggle.com/datasets/ammaraahmad/immigration-to-canada |
| 9 | Given flight data for January 2019 and 2020 showing on-time performance:<br>• Visualize and compare the statistics of flight distances for both years using appropriate plotting techniques.<br>• Visualize the number of flights with destination airport IDs 11778 and 11267 using a bar plot or bar chart.<br>• Create a sunburst plot for both years to depict the differences in flight data.<br>https://www.kaggle.com/datasets/divyansh22/flight-delay-prediction |
| 10 | Given flight data for January 2019 and 2020 showing on-time performance:<br>• Visualize and compare the statistics of flight distances for both years using appropriate plotting techniques.<br>• Visualize the number of flights with destination airport IDs 11778 and 11267 using a bar plot or bar chart.<br>• Create a sunburst plot for both years to depict the differences in flight data.<br>https://www.kaggle.com/datasets/barkhaverma/placement-data-full-class |

**Text Books:**

1. Igor Milovanovic, DimitryFoures, and Giuseppe Vettigli, "Python Data Visualization Cookbook"
2. Scott Murray "Interactive Data Visualization for the Web"
3. Kyran Dale "Data Visualization with Python and JavaScript"

**Reference Books:**

1. Al Sweigart, "Automate the Boring Stuff with Python",1stEdition, No Starch Press, 2015. (Available under CC-BY-NC-SA license at https://automatetheboringstuff.com/)
2. Reema Thareja "Python Programming Using Problem Solving Approach" Oxford University Press.
3. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2nd Edition, Green Tea Press, 2015. (Available under CC-BY-NC license at http://greenteapress.com/thinkpython2/thinkpython2.pdf)

**3. Teaching-Learning Process Strategies**

| S/L | TLP Strategies: | Description |
|---|---|---|
| 1. | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2. | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 3. | Higher Order Thinking (HOTS) Questions: | Pose HOTS questions to stimulate critical thinking related to each competency. |
| 4. | Problem-Based Learning (PBL) | Implement PBL to enhance analytical skills and practical application of competencies |
| 5. | Multiple Representations | Introduce topics in various representations to reinforce competencies |
| 6. | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 7. | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 8. | Programming Assignments | Assign programming tasks to reinforce practical skills associated with competencies. |

**4. Assessment Details (both CIE and SEE)**

**Test Marks distribution for the Practical Course**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Write-up, Conduction, Result and Procedure/Algorithm/Flowchart | 60% | 60 |
| 2 | Viva-Voce | 40% | 40 |
| **Total** | | **100%** | **100** |

**Final CIE in Practical Course including AEC:**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Scaled Down marks of conduction, record/journal | 60% of the maximum | 30 |
| 2 | Scaled-down marks of the test | 40% of the maximum | 20 |
| **Total** | | **100%** | **50** |

**SEE for the Practical Course:**

1. SEE marks for a practical course shall be 50 marks.
2. The practical course is evaluated for 100 marks, and the scored marks shall be scaled down to 50 marks.
3. A change of experiment/program is allowed only once, and 20% of the marks allotted to the procedure/write-up part will be zero.
4. The duration of SEE shall be 3 hours.

**Marks distribution for the Practical Course**

| SL. No. | Description | % Marks | Marks |
|---|---|---|---|
| 1 | Write-up, Procedure | 20% | 20 |
| 2 | Conduction and result | 60% | 60 |
| 3 | Viva-Voce | 20% | 20 |
| **Total** | | **100%** | **100** |

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS307A.1** | Demonstrate the use of IDLE or PyCharm IDE to create Python Applications. |
| **M25BCS307A.2** | Use Python programming constructs to develop programs for solving real-world problems |
| **M25BCS307A.3** | Use Matplotlib for drawing different Plots |
| **M25BCS307A.4** | Demonstrate working with Seaborn, Bokeh for visualization |
| **M25BCS307A.5** | Use Plotly for drawing Time Series and Maps |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS307A.1** | 3 | - | - | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS307A.2** | - | 3 | - | - | - | - | - | - | - | - | - | - | - |
| **M25BCS307A.3** | - | - | 3 | - | - | - | - | - | - | - | - | 2 | 2 |
| **M25BCS307A.4** | - | - | - | - | 3 | - | - | - | - | - | - | - | 2 |
| **M25BCS307A.5** | - | - | - | - | 3 | - | - | - | - | - | - | 2 | - |
| **M25BCS307A** | **3** | **3** | **3** | **-** | **3** | - | - | - | - | - | - | **2** | **2** |

**6.Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| **1** | **SDG 3:**Good Health and Well-Being | Visualizing health data (e.g., disease spread, vaccination rates) helps in early detection and policy planning.Python libraries like **Matplotlib** and **Seaborn** are used to create health dashboards and reports. |
| **2** | **SDG 7:**Affordable and Clean Energy | Visualizations of energy consumption patterns help promote energy-saving habits and clean energy policies.Tools like Plotly can display real-time energy data from IoT devices or sensors. |
| **3** | **SDG 9:**Industry, Innovation, and Infrastructure | Helps in analyzing industrial performance through data dashboards and trend graphs.Supports innovation via data-driven decisions in smart infrastructure and automation. |

| 3rd Semester | Ability Enhancement Course (AE)<br>PHP PROGRAMMING | M25BCS307B |
|---|---|---|

**1. Prerequisites**

| S/L | Proficiency | Prerequisites |
|---|---|---|
| 1 | Basic Computer Skills | Familiarity with using a **computer system**, file handling, and basic navigation in an operating system. |
| 2 | HTML & CSS | Understanding of HTML elements (forms, inputs, tags) and basic CSS for structuring and styling web pages. |
| 3 | Basic Programming concepts | It's beneficial to have a solid understanding of basic programming concepts such as variables, data types, loops, conditional statements, functions, and arrays. These concepts are fundamental to programming in any language, including PHP |
| 4 | HTTP and Web Servers | Understanding how the web works, including concepts like HTTP (Hypertext Transfer Protocol) and how web servers operate, will give you a better understanding of how PHP scripts interact with web browsers and servers. |
| 5 | Database Concepts | PHP is often used in conjunction with databases to create dynamic web applications. Familiarize yourself with concepts like database design, SQL (Structured Query Language), and basic database operations such as querying, inserting, updating, and deleting data. |

**2. Syllabus**

| PHP PROGRAMMING<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS307B** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L: T: P:TW:SL) | **0:0:32:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course objectives:** This course will enable students to:

1. To introduce the PHP syntax, elements, and control structures.
2. To make use of PHP Functions and File handling.
3. To illustrate the concept of PHP arrays and OOPs.

| Sl. No | Experiments |
|---|---|
| 1 | A junior web developer is working at a tech company that is building a portal for school-level programming practice. One of the modules on the portal allows students to input three numbers and find the largest among them using PHP. Create a PHP web page using functions to compare three integers and print the largest number. |
| 2 | A backend module is being developed for a mathematics learning application targeted at school-level students. One of the modules is designed to help students understand the concept of factorials. A PHP function is required that calculates the factorial of a non-negative integer. The function should accept the number as an argument and return the result. **Create a PHP script using a user-defined function to calculate and display the factorial of a given non-negative integer.** |
| 3 | A simple tool is being developed for a language learning platform where users can input words or phrases and explore string manipulation concepts. One of the features requires reversing a string entered by the user. Create a PHP page where a user can enter a string, and upon submission, the page should display the reverse of the entered string. |
| 4 | A project is being developed to help students explore interesting properties of words and phrases. One of the required features is to check whether a given string is a palindrome. A palindrome is a word or phrase that reads the same backward as forward (e.g., “madam”, “racecar”, “121”).Create a PHP script that accepts a string from the user and displays whether the string is a palindrome or not. |
| 5 | A text-cleaning utility is being developed for a content moderation system. The system requires removal of unnecessary spaces from user inputs before further processing. Create a PHP script that accepts a string and removes all whitespaces from it. For **Example:**Input: 'The quick"" brown fox'Expected Output: Thequick""brownfox |
| 6 | A basic mathematics practice tool is being developed to help school students understand number patterns. One of the features requires calculating the sum of the first *n* odd numbers (e.g., 1, 3, 5, ...).Create a PHP script that accepts a positive integer *n* from the user and displays the sum of the first *n* odd numbers. |
| 7 | A secure login system is being developed for an online student portal. The login page must |

| | |
|---|---|
| | prompt users to enter their username and password. Upon submission, the entered credentials should be validated against user records stored in a MySQL database.If the credentials match, a personalized welcome message should be displayed. Otherwise, an error message stating "Invalid Username or Password" should be shown. |
| 8 | A basic search feature is being developed for a blogging platform. The functionality should allow users to verify whether a given keyword or phrase exists within the content of a blog post. Create a PHP script that accepts a main string and a substring from the user and checks if the substring is present within the main string. If found, a confirmation message should be displayed; otherwise, an appropriate message indicating that the substring is not found should be shown. |
| 9 | A personalized dashboard is being developed for a greeting card website. One of the key features is a "Birthday Countdown," which displays the number of days remaining until the user's next birthday. Create a PHP script that accepts the user's birth date (day and month) as input and calculates the number of days remaining from the current date to the next occurrence of their birthday. The countdown should dynamically update based on the system's current date. |
| 10 | A registration form is being developed for a web application in which users are required to enter their email addresses. To ensure that the input is in the correct format, email validation must be implemented. Create a PHP script that accepts an email address from the user and verifies its validity using appropriate PHP functions. If the email is valid, a success message should be displayed; otherwise, an error message should be shown. |
| 11 | A multilingual greeting system is to be designed for a tourism website. The system should greet users in their chosen language. Create an HTML form containing a dropdown list with language options such as English, Hindi, French, Spanish, and German. Upon form submission, use a PHP script with a switch-case structure to display a greeting ("Hello") in the selected language. |
| 12 | A module is being developed for a programming learning portal to demonstrate the concept of recursion. One of the tasks involves generating the Fibonacci series using a recursive function. Create a PHP script that accepts a number n as input and displays the first n terms of the Fibonacci series by implementing recursion. |
| 13 | A content-editing tool is being developed to assist users in auto-correcting or updating specific words in a given text. One of the features requires replacing only the first occurrence of a specific word within a paragraph. Create a PHP script that accepts a string input and replaces the first occurrence of the word **"the"** with **"That"**. For example, given the input:"the quick brown fox jumps over the lazy dog."The output should be:"That quick brown fox jumps over the lazy dog." |

**Text Books:**

1. DEVID R BROOKS, "Programming in HTML and PHP Coding for Scientists and Engineers",Springer International Publishing AG 2017 .
2. Steven Holzner, "PHP: The Complete Reference Paperback", McGraw Hill Education (India), 2007.

**Reference Books:**

1. David Sklar, Adam Trachtenberg, "PHP Cookbook: Solutions & Examples for PHP Programmers", 2014.
2. Programming PHP, RasmusLerdorf, Kevin Tatroe, Peter MacIntyre,"O'Reilly Media, Inc.", 28 Apr 2006.

**3. Teaching-Learning Process Strategies**

| S/L | TLP Strategies: | Description |
|---|---|---|
| 1 | Pre-Lab Sessions | Instructor introduces foundational **theoretical concepts** related to PHP such as:<br>• Client-server architecture<br>• PHP syntax and data types<br>• Form handling and variables |
| 2 | Pre-Lab Readings | Students are assigned **textbook chapters**, online tutorials, or reference documentation (e.g., php.net) to familiarize with topics **before lab sessions**. |
| 3 | Hands-On Coding Practice | Students write PHP scripts to perform specific tasks such as:<br>• Displaying output using echo<br>• Handling forms<br>• Using conditionals, loops, and arrays<br>• Connecting to MySQL |
| 4 | Demonstration & Guidance | Instructor **demonstrates coding** examples, explains output, and assists students with:<br>• Syntax errors<br>• Logic building |

| | | |
|---|---|---|
| | | • Troubleshooting during development |
| 5 | Lab Reports, conclusion & Inference | Students submit structured **lab reports** including:<br>• Objective of the task<br>• Source code and explanation<br>• Screenshots of output<br>• Inferences and challenges faced |

**4. Assessment Details (both CIE and SEE)**

**Test Marks distribution for Ability Enhancement (Practical) Course**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Write-up, Conduction, Result and Procedure/Algorithm/Flowchart | 60% | 60 |
| 2 | Viva-Voce | 40% | 40 |
| **Total** | | **100%** | **100** |

**Final CIE in Practical Course including AEC:**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Scaled Down marks of conduction, record/journal | 60% of the maximum | 30 |
| 2 | Scaled-down marks of the test | 40% of the maximum | 20 |
| **Total** | | **100%** | **50** |

**SEE for the Practical Course:**

1. SEE marks for a practical course shall be 50 marks.
2. The practical course is evaluated for 100 marks, and the scored marks shall be scaled down to 50 marks.
3. A change of experiment/program is allowed only once, and 20% of the marks allotted to the procedure/write-up part will be zero.
4. The duration of SEE shall be 3 hours.

**Marks distribution for the Practical Course**

| SL. No. | Description | % Marks | Marks |
|---|---|---|---|
| 1 | Write-up, Procedure | 20% | 20 |
| 2 | Conduction and result | 60% | 60 |
| 3 | Viva-Voce | 20% | 20 |
| **Total** | | **100%** | **100** |

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS307B.1** | Apply basic concepts of PHP to develop web program |
| **M25BCS307B.2** | Develop programs in PHP involving control structures |
| **M25BCS307B.3** | Develop programs to handle structured data (object) and data items (array) |
| **M25BCS307B.4** | Develop programs to access and manipulate contents of files |
| **M25BCS307B.5** | Use super-global arrays and regular expressions to solve real world problems. |

**CO-PO-PSO Mapping**

| COs/Pos | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS307B.1** | 3 | - | - | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS307B.2** | - | 3 | - | - | - | - | - | - | - | - | - | 2 | - |
| **M25BCS307B.3** | - | - | 3 | - | - | - | - | - | - | - | - | - | 2 |
| **M25BCS307B.4** | - | - | - | - | 3 | - | - | - | - | - | - | 2 | - |
| **M25BCS307B.5** | - | - | - | - | 3 | - | - | - | - | - | - | - | 2 |
| **M25BCS307B** | **3** | **3** | **3** | - | **3** | - | - | - | - | - | - | **2** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | SDG-4: Quality Education | PHP programming introduces students to dynamic web development, enhancing digital literacy and employable skills, thereby promoting inclusive and equitable quality education. |

| | | |
|---|---|---|
| 2 | SDG 8: Decent Work and Economic Growth | By learning industry-relevant web technologies like PHP, students are better equipped for IT job markets, contributing to economic productivity and decent work opportunities. |
| 3 | SDG 9: Industry, Innovation, and Infrastructure | PHP is widely used in developing scalable web applications and digital infrastructure. Mastery of such technologies fosters innovation and supports robust digital industry development. |

| 3rd Semester | Ability Enhancement Course (AE)<br>PROJECT MANAGEMENT WITH REPORT WRITING | M25BCS307C |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic Computer Skills | Familiarity with File Management: Ability to manage files and folders on a computer.<br>Basic Software Skills: Comfortable using web browsers and common software applications. Typing Proficiency: Ability to type efficiently and use keyboard shortcuts. |
| 2 | Basic Understanding of Document Preparation | Word Processing: Basic knowledge of using word processors like Microsoft Word or Google Docs.<br>Formatting Documents: Understanding of basic document formatting, such as setting margins, headings, and paragraphs. |
| 3 | Fundamental Mathematical Knowledge | Mathematical Notation: Familiarity with basic mathematical symbols and notation.<br>Basic Algebra and Calculus: Understanding of fundamental concepts in algebra and calculus, as these are often needed for typesetting equations. |
| 4 | Introduction to Programming or Markup Languages | Basic Programming Concepts: Familiarity with basic programming or scripting concepts can be beneficial but is not mandatory.<br>Markup Languages: Understanding of HTML, Markdown, or other markup languages can be helpful for grasping LaTeX syntax. |
| 5 | Basic Knowledge of Bibliographies and Citations | Understanding Citations: Familiarity with the purpose and format of citations and references. Bibliography Management: Basic knowledge of bibliography management tools like EndNote, Zotero, or Mendeley |
| 6 | English Proficiency | Reading and Writing: Proficiency in reading and writing English, as most of the documentation and tutorials for Overleaf and LaTeX are in English |

**2. Syllabus**

| PROJECT MANAGEMENT WITH REPORT WRITING<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS307C** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L: T: P:TW:SL) | **0:0:32:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course Objectives:**

1. To, learn what stress and strain are and how materials behave under pressure.
2. To analyzing structures figure out how different structures respond to different types of forces.
3. To Calculating Forces and learn how to calculate the forces and moments acting on beams, shafts, and columns.
4. Testing Materials to understand how materials respond to stress and when they might break.
5. Designing Structures: Use stress analysis to design strong and safe structures.
6. Get familiar with computer programs that help with stress analysis and design.

**Module -1**

**Creating a document in Overleaf**

Uploading a project, copying a project, Creating a project from a template, Using the Overleaf project menu, Including images in Overleaf, Exporting your work from Overleaf, Working offline in Overleaf, Using Track Changes in Overleaf, Using bibliographies in Overleaf, Sharing your work with others, Using the History feature.

**Module -2**

**Mathematics**

Mathematical expressions, Subscripts and superscripts, Brackets and Parentheses, Matrices, Fractions and Binomials, aligning equations, Operators, spacing in math mode, Integrals, sums and limits, display style in math mode, List of Greek letters and math symbols, Mathematical fonts, Using the Symbol Palette in Overleaf.

**Module -3**

| |
|---|
| **Figures and tables**<br>Inserting Images, Tables, Positioning Images and Tables, Lists of Tables and Figures, Drawing Diagrams, Directly in LaTeX<br>**References and Citations**<br>Bibliography management with bibtex, Bibliography management with natbib, Bibliography management with biblatex, Bibtex bibliography styles, Natbib bibliography styles, Natbib citation styles, Biblatex bibliography styles, Biblatex citation styles. |
| **Module -4** |
| **Languages:** Multilingual typesetting on Overleaf using polyglossia and fontspec, Multilingual typesetting on Overleaf using babel and fontspec, international language support, Quotations and quotation marks.<br>**Document structure** Sections and chapters, Table of contents, Cross referencing sections, equations and floats, Indices, Glossaries, Nomenclatures, Multi-file LaTeX projects, Hyperlinks. |
| **Module -5** |
| **Formatting** Lengths in LaTeX, Headers and footers, Page numbering, Paragraph formatting, Line breaks and blank spaces, Text alignment, Page size and margins, Single sided and double sided documents, Multiple columns, Counters, Footnotes, Margin notes<br>**Fonts** Font sizes, families, and styles, Font typefaces, Supporting modern fonts with LaTeX<br>**Presentations** Beamer, Power dot, Posters,<br>**Commands** Commands Environments<br>**Field specific** Theorems and proofs, Chemistry formulae, Feynman diagrams, Molecular orbital diagrams Chess notation. |
| **Reference:** https://www.overleaf.com/learn (details for all individual topics are available at this link) |

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Sequential Learning | Begin with basic concepts like creating a document, uploading a project, and copying a project. |
| 2 | Step-by-Step Tutorials | Gradually introduce more complex topics like mathematical expressions, figures, tables, and bibliographies. |
| 3 | Interactive Sessions | Conduct live coding sessions where students follow along on their own Overleaf accounts. |
| 4 | Assignments | Give practical assignments requiring the use of specific Overleaf features (e.g., creating a document with a bibliography and figures) |
| 5 | Projects | Have students work on larger projects that incorporate multiple Overleaf and LaTeX features. |
| 6 | Project-Based Learning | Engage in projects that require comprehensive use of LaTeX and Overleaf. |

4. **Assessment Details (both CIE and SEE)**

**Test Marks distribution for the Ability Enhancement (Practical) Course**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Write-up, Conduction, Result and Procedure/Algorithm/Flowchart | 60% | 60 |
| 2 | Viva-Voce | 40% | 40 |
| **Total** | | **100%** | **100** |

**Final CIE in Practical Course including AE Course:**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Scaled Down marks of conduction, record/journal | 60% of the maximum | 30 |
| 2 | Scaled-down marks of the test | 40% of the maximum | 20 |
| **Total** | | **100%** | **50** |

**SEE for the Practical Course:**

1. SEE marks for a practical course shall be 50 marks.
2. The practical course is evaluated for 100 marks, and the scored marks shall be scaled down to 50 marks.
3. A change of experiment/program is allowed only once, and 20% of the marks allotted to the procedure/write-up part will be zero.
4. The duration of SEE shall be 3 hours.

**Marks distribution for the Practical Course**

| SL. No. | Description | % Marks | Marks |
|---|---|---|---|
| 1 | Write-up, Procedure | 20% | 20 |
| 2 | Conduction and result | 60% | 60 |
| 3 | Viva-Voce | 20% | 20 |
| **Total** | | **100%** | **100** |

## 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS307C.1** | Apply knowledge to create, manage, and export documents in Overleaf, including uploading existing projects, utilizing templates, and working offline. |
| **M25BCS307C.2** | Analyze and construct complex mathematical content using LaTeX, including expressions, equations, matrices, and symbols |
| **M25BCS307C.3** | Design lists of tables and figures and develop diagrams directly within LaTeX to enhance the visual structure of documents. |
| **M25BCS307C.4** | Evaluate and implement effective bibliography and citation management using bibtex, natbib, and biblatex. Critically apply different bibliography and citation styles to ensure accurate and consistent referencing in academic documents. |
| **M25BCS307C.5** | Synthesize multilingual typesetting techniques using polyglossia, babel, and fontspec. Organize and structure documents with sections, chapters, cross-referencing, indices, glossaries, and hyperlinks. |

**CO-PO-PSO Mapping**

| COs/Pos | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS307C.1** | 3 | 2 | 2 | - | 3 | - | - | - | 2 | 3 | 2 | - | - |
| **M25BCS307C.2** | 3 | 3 | 2 | 3 | 3 | - | - | - | - | 2 | 2 | - | 2 |
| **M25BCS307C.3** | 3 | 2 | 3 | - | 3 | - | - | - | 2 | 2 | 3 | 2 | - |
| **M25BCS307C.4** | 3 | 3 | 2 | 3 | 3 | - | - | - | 2 | 2 | 2 | - | - |
| **M25BCS307C.5** | 3 | 2 | 3 | 2 | 3 | - | - | - | 2 | 3 | 3 | 2 | 2 |
| **M25BCS307C** | **3** | **2.4** | **2.4** | **2.6** | **3** | **-** | **-** | **-** | **2** | **2.6** | **2.4** | **2** | **2** |

## 6. Mapping to Sustainable Development Goals (SDG):

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4**: Quality Education | The course enhances students' ability to plan, execute, and present technical projects effectively, promoting lifelong learning skills and communication, thereby contributing to SDG 4. |
| 2 | **SDG 8**: Decent Work and Economic Growth | By equipping students with skills in project planning, time management, and documentation, the course prepares them for the workforce and supports productivity and innovation, aligning with SDG 8. |
| 3 | **SDG 9:** Industry, Innovation and Infrastructure | Project management methodologies introduced in the course foster structured problem-solving and innovation in engineering domains, aiding in sustainable industrial and infrastructural development (SDG 9). |

| 3rd Semester | Ability Enhancement Course (AE)<br>DATA ANALYTICS WITH EXCEL | M25BCS307D |
|---|---|---|

**1. Prerequisites**

| S/L | Proficiency | Prerequisites |
|---|---|---|
| 1 | Basic Excel skills | You should be comfortable with navigating Excel, entering data, formatting cells, using formulas and functions, and creating basic charts and tables. |
| 2 | Understanding of data types | Knowing the different types of data (e.g., numerical, text, date/time) and how Excel handles them is essential for data analysis |
| 3 | Knowledge of Excel functions | Understanding commonly used functions like SUM, AVERAGE, COUNT, IF, VLOOKUP, INDEX/MATCH, and pivot tables will be beneficial for data analysis tasks |
| 4 | Data cleaning skills | Being able to clean and prepare data for analysis, including dealing with missing values, duplicates, and inconsistencies, is crucial for accurate analysis. |
| 5 | Basic statistical concepts | Familiarity with basic statistical concepts like mean, median, mode, standard deviation, and correlation will help you interpret and analyze data effectively |
| 6 | Pivot tables | Knowing how to create and manipulate pivot tables is essential for summarizing and analyzing large datasets in Excel |
| 7 | Data visualization | Understanding how to create effective data visualizations such as charts and graphs to communicate insights from your analysis. |
| 8 | Data analysis tools | Familiarity with Excels data analysis tools, such as regression analysis, scenario analysis, and data modeling, can enhance your analytical capabilities. |

**2. Syllabus**

| DATA ANALYTICS WITH EXCEL<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS307D** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **0:0:32:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course objectives:** This course will enable students to:

1. To Apply analysis techniques to datasets in Excel
2. Learn how to use Pivot Tables and Pivot Charts to streamline your workflow in Excel
3. Understand and identify the principles of data analysis
4. Become adept at using Excel functions and techniques for analysis
5. Build presentation ready dashboards in Excel

| Sl. No | Experiments |
|---|---|
| 1 | An assignment has been given to maintain a student database for a college using Excel. The database should include fields such as student USN, name, department, course, and semester. As new data is received, additional rows and columns must be inserted. The Drag & Fill feature is to be used for entering repetitive information efficiently, and aggregate functions should be applied to analyze the student data. How can this Excel sheet be organized and managed effectively to accomplish these tasks? |
| 2 | A task has been assigned to create and manage a student database in Excel with fields including student USN, name, department, course, and semester. The responsibilities include importing existing student data from an external source, entering new records, manipulating the data as required, and using sorting and filtering tools to organize and analyze the information efficiently. What approach should be taken to ensure the database is managed accurately and effectively? |
| 3 | Responsibility has been assigned to maintain a student database in Excel with fields such as student USN, name, department, course, and semester. To ensure data accuracy, data validation must be implemented for appropriate entries. Additionally, the data needs to be analyzed using pivot tables and presented through pivot charts. How can these features be set up and managed to maintain data integrity and generate meaningful reports? |
| 4 | A task has been assigned to create and analyze a student database in Excel containing fields such as student USN, name, department, course, and semester. Conditional formatting should be applied to highlight specific data trends, What-If Analysis used to explore various scenarios, data |

| | |
|---|---|
| | tables utilized for structured management, and charts or graphs created to visually represent the information. How can these Excel features be implemented effectively to analyze and present the student data? |
| 5 | A worksheet containing employee details—including Empno, Ename, Basic Pay, Travelling Allowance, Dearness Allowance, House Rent Allowance, Income Tax, Provident Fund, and Net Pay—has been provided, but the data includes inconsistencies like extra spaces and inconsistent text casing. How can Excel functions such as UPPER, LOWER, TRIM, and CONCATENATE be applied to clean and standardize employee names and combine relevant fields to enhance data presentation? |
| 6 | A worksheet contains employee details including Empno, Ename, Basic Pay, Travelling Allowance, Dearness Allowance, House Rent Allowance, Income Tax, Provident Fund, and Net Pay, along with various date and time entries in inconsistent formats. How can Excel functions such as DATEVALUE, DATEADD, DATEDIF, and TIMEVALUE be used to clean, convert, and analyze the date and time data to ensure accurate reporting? |
| 7 | An employee payroll worksheet is being managed with fields such as Empno, Ename, Basic Pay, Travelling Allowance, Dearness Allowance, House Rent Allowance, Income Tax, Provident Fund, and Net Pay. How can conditional formatting be applied to highlight important data patterns, parse relevant information, and visually differentiate key values to enhance data analysis and support effective decision-making? |
| 8 | An employee payroll worksheet is being managed with fields such as Empno, Ename, Basic Pay, Travelling Allowance, Dearness Allowance, House Rent Allowance, Income Tax, Provident Fund, and Net Pay. How can conditional formatting be applied to highlight important data patterns, parse relevant information, and visually differentiate key values to enhance data analysis and support effective decision-making? |
| 9 | A company wants to process its employee payroll for the month. Each employee has a basic pay (BP), and based on that, the company provides various allowances: Travelling Allowance (TA) as 10% of BP, Dearness Allowance (DA) as 15% of BP, and House Rent Allowance (HRA) as 20% of BP. From the gross pay (BP + TA + DA + HRA), the company deducts Income Tax (IT) at 12% and Provident Fund (PF) at 8%. You need to create a worksheet to calculate the Net Pay (NP) for each employee after deductions, analyze the payroll data with charts, and generate a report summarizing employee earnings and deductions. |
| 10 | A retail store wants to manage its inventory by tracking products with details including product code, name, type, MRP, and purchase date. The store offers discounts on products, so the cost price after applying the discount percentage needs to be calculated. You are required to create a worksheet to record this data, calculate the discounted cost price, analyze the inventory trends using charts, and prepare a report on product categories, pricing, and purchase patterns. |
| 11 | A merchandise store wants to analyze its sales data including order details, customer demographics, product categories, and sales channels. You need to create a worksheet capturing all relevant data, use formulas to segregate product categories, and perform a comparative study of sales by gender, age group, platform, and city using pivot tables and various charts to identify trends and insights. |
| 12 | A company wants to automate the generation of sales reports and presentations by using Auto filter to filter relevant data and macros to streamline repetitive tasks. You are required to create a workbook that uses Auto filter and macros to efficiently produce customized reports and presentation-ready summaries. |

**Text Books:**

1. M 1. Berk& Carey - Data Analysis with Microsoft® Excel: Updated for Office 2007®, Third Edition, © 2010 Brooks/Cole, Cengage Learning, ISBN-13: 978-0-495-39178-4
2. Wayne L. Winston - Microsoft Excel 2019: Data Analysis and Business Modeling, PHI, ISBN: 9789389347180

**Reference Books:**

1. Aryan Gupta - Data Analysis in Excel: The Best Guide. (https://www.simplilearn.com/tutorials/excel-tutorial/data-analysis-excel)Instrumentation, Measurement and Analysis B C Nakra, K KChaudhry McGraw–Hill 4th Edition
2. Comprehensive & Easy Guide to Learn Advanced MS Excel: By Naveen Mishra

**3. Teaching-Learning Process Strategies**

| S/L | TLP Strategies: | Description |
|---|---|---|
| 1 | Pre-Lab Sessions | Lecturer delivers discussions on the theoretical concepts relevant to the specific lab experiment. This could cover topics like: |

| | | |
|---|---|---|
| | | • The principles of measurement, Uncertainty and errors in measurement, Operation and functionalities of the measuring instruments used in the experiment and Calibration methods for instruments. |
| 2 | Pre-Lab Readings | Students are assigned relevant readings from textbooks or lab manuals, to make their understanding before entering the lab. |
| 3 | Experimentation | Students work in groups or individually to perform the assigned experiments as outlined in the lab manual. This involves:<br>• Setting up the measuring instruments according to the procedure.<br>• Taking measurements of the designated objects or parameters.<br>• Recording data meticulously.<br>• Performing calculations based on the collected data.<br>• Analyzing the results and interpreting their meaning. |
| 4 | Demonstration & Guidance | Lecturer providing guidance and assistance to students as needed. This could involve:<br>• Demonstrating proper use of the instruments.<br>• Addressing questions and troubleshooting any issues encountered.<br>• Ensuring students adhere to safety protocols. |
| 5 | Lab Reports, conclusion & Inference | Students prepare reports summarizing their lab experience. These reports typically include:<br>• Objectives of the experiment.<br>• Description of the procedure followed.<br>• Recorded data in tables or graphs.<br>• Calculations performed and analyzed results.<br>• Discussion of observations, sources of error, and conclusions drawn |

**4. Assessment Details (both CIE and SEE)**

**Test Marks distribution for the Ability Enhancement (Practical) Course**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Write-up, Conduction, Result and Procedure/Algorithm/Flowchart | 60% | 60 |
| 2 | Viva-Voce | 40% | 40 |
| **Total** | | **100%** | **100** |

**Final CIE in Practical Course including AE Course:**

| Sl. No. | Description | % Marks | In Marks |
|---|---|---|---|
| 1 | Scaled Down marks of conduction, record/journal | 60% of the maximum | 30 |
| 2 | Scaled-down marks of the test | 40% of the maximum | 20 |
| **Total** | | **100%** | **50** |

**SEE for the Practical Course:**

1. SEE marks for a practical course shall be 50 marks.
2. The practical course is evaluated for 100 marks, and the scored marks shall be scaled down to 50 marks.
3. A change of experiment/program is allowed only once, and 20% of the marks allotted to the procedure/write-up part will be zero.
4. The duration of SEE shall be 3 hours.

**Marks distribution for the Practical Course**

| SL. No. | Description | % Marks | Marks |
|---|---|---|---|
| 1 | Write-up, Procedure | 20% | 20 |
| 2 | Conduction and result | 60% | 60 |
| 3 | Viva-Voce | 20% | 20 |
| **Total** | | **100%** | **100** |

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS307D.1** | Employing productivity tools and advanced functions to help with worksheet creation and development. |
| **M25BCS307D.2** | Utilizing outline pivot tables, and manipulating data listings. |
| **M25BCS307D.3** | Use consolidation to compile and report findings from multiple worksheets, |

| | |
|---|---|
| **M25BCS307D.4** | Apply macros and auto filters to solve the real-world scenario |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS307D.1** | 3 | 3 | 2 | - | 3 | - | - | - | - | - | - | 2 | 2 |
| **M25BCS307D.2** | 3 | 3 | 2 | - | 3 | - | - | - | - | - | - | 2 | - |
| **M25BCS307D.3** | 3 | 3 | 2 | - | 3 | - | - | - | - | - | - | - | - |
| **M25BCS307D.4** | 3 | 3 | 2 | - | 3 | - | - | - | - | - | - | 3 | 2 |
| **M25BCS307D** | **3** | **3** | **2** | **-** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **2.33** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4 -** Quality Education | Excel-based analytics helps in better understanding of data interpretation, enhancing digital literacy. |
| 2 | **SDG 8** - Decent Work and Economic Growth | Excel skills are in high demand in job markets; they support business decisions and promote productivity. |
| 3 | **SDG 9** - Industry, Innovation, and Infrastructure | Using Excel for data-driven solutions fosters innovation in small-scale industries and organizations. |
| 4 | **SDG 12** - Responsible Consumption and Production | Excel analytics can track and optimize resource usage, reduce waste, and promote sustainable practices. |

| 3rd Semester | Ability Enhancement Course (AE)<br>BIOTECHNOLOGY | M25BCS308A |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Biotechnology | The broad field involving the use of biological systems, organisms, or derivatives to develop useful technologies for industrial, agricultural, and medical applications. |
| 2 | Genetic Engineering | A branch of biotechnology focused on modifying genetic material (DNA) to alter or enhance organisms for specific purposes using tools like recombinant DNA, PCR, and cloning. |
| 3 | Bioprocess Technology | Engineering principles applied to biological processes, including microbial fermentation, bioreactor design, and downstream processing to produce products at industrial scale. |
| 4 | Biomedical Applications | Use of biotechnology in healthcare for developing biomaterials, tissue engineering, biosensors, and drug delivery systems to improve diagnosis and treatment. |

**2. Syllabus**

**BIOTECHNOLOGY**
**SEMESTER – III**

| | | | |
|---|---|---|---|
| Course Code | **M25BCS308A** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem (L:T: P:TW:SL) | **32:0:0:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **01** |

**Course Objectives:**

1. Understand the scope and applications of biotechnology in various engineering and industrial sectors.
2. To Learn basics of cell biology, genetics, and molecular biology for biotechnology applications.
3. Explore genetic engineering tools and microbial techniques used in biotechnology processes.
4. Study bioprocess principles including fermentation, bioreactor operation, and downstream processing techniques.
5. Understand environmental biotechnology applications like bioremediation, waste management, and sustainable development.
6. Discover biotechnology integration in biomedical devices, diagnostics, and modern healthcare innovations.

**Module -1**

**Introduction to Biotechnology and Its Engineering Applications:** Definition, scope, and branches of biotechnology.
**Role of biotechnology in engineering fields:** biomedical, agricultural, environmental, industrial Biotech industry and market overview Ethical, legal, and social issues in biotechnology**.**
**Engineering Focus:** Understanding biotech impact on engineering sectors and technology development.
**Text Book1: chapter 1** **10 Hours**

**Module -2**

**Cell Biology and Genetic Engineering Basics for Engineers:** Cell structure and functions relevant to biotech processes
**Basics of genetics:** DNA, genes, chromosomes
**Principles of genetic engineering:** cloning, recombinant DNA technology
**Tools and techniques:** restriction enzymes, vectors, PCR
**Engineering Focus:** Core biological concepts supporting bioengineering and genetic manipulation techniques
**Text Book: chapter 1** **10 Hours**

**Module -3**

**Microbial Technology and Bioprocess Engineering:** Microbial classification and growth kinetics.
**Fermentation technology:** batch, fed-batch, continuous culture, Design and operation of bioreactors, Downstream processing and product recovery
**Engineering Focus:** Application of microbiology in process engineering, reactor design, and scale-up.
**Text Book: chapter 2** **10 Hours**

**TEXT BOOKS:**

1. P.K. Gupta, Element of Biotechnology, 4th Edition, Rastogi Publications, 2018.
2. Arun Kumar, Biotechnology for Engineers, 2nd Edition, PHI Learning Pvt. Ltd., 2020

**VIDEO LINKS:**

3. https://www.youtube.com/watch?v=IXWYsgR9Y5Y
4. https://www.youtube.com/watch?v=tPbII9E6qCA
5. https://www.youtube.com/watch?v=Sn0bOX5Hau

### 3. Teaching-Learning Process Strategies

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of Biotechnology concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |

### 4. Assessment Details (both CIE and SEE)

**Continuous Internal Evaluation:**

**CIE Split up for Ability Enhancement Course (AE)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B) | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

1. The CIE question paper shall have MCQ set for 25 questions, each carrying one mark.
2. The average internal assessment shall be the average of the 2 best marks in the tests conducted

**Semester End Examination:**

1. The SEE question paper shall have MCQ set for 50 questions, each carrying one mark.
2. The time duration for SEE is one hour.

### 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS308A.1** | Understand and Apply the fundamental concepts, scope, and branches of biotechnology. |
| **M25BCS308A.2** | Apply basic knowledge of cell biology and genetics in engineering applications. |
| **M25BCS308A.3** | Analyzemicrobial growth and design suitable bioprocess techniques. |
| **M25BCS308A.4** | Design biotechnological solutions for environmental issues like pollution and waste management. |
| **M25BCS308A.5** | Evaluate applications of biotechnology in biomedical engineering and recent innovations. |

**CO-PO-PSO Mapping**

| COs/Pos | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS308A.1** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS308A.2** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS308A.3** | - | 3 | - | - | - | - | - | - | - | - | - | - | - |
| **M25BCS308A 4** | - | - | 3 | - | - | - | - | - | - | - | - | - | - |
| **M25BCS308A.5** | - | - | - | 3 | - | - | - | - | - | - | - | - | - |
| **M25BCS308A** | **3** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **-** |

**6. Mapping to Sustainable Development Goals (SDG)**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 3**: Good Health and Well-Being | Biomedical Engineering, biosensors, drug delivery systems |
| 2 | **SDG 6**: Clean Water and Sanitation | Wastewater treatment, biofilters, microbial degradation |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure, | Biotech innovations, fermentation, bioprocess design |

| 3rd Semester | Ability Enhancement Course (AE)<br>AI FOR TECHNOLOGY | M25BCS308B |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Health care | Basic human anatomy and physiology, clinical terminology, medical work flow. |
| 2 | Mathematics and Statistics | Linear algebra, calculus, probability, hypothesis testing, regression. |
| 3 | Programming | Python Fundamentals, Data Structures |
| 4 | Domain knowledge in Healthcare | Familiarity with medical data types, ethical and legal aspects. |

**2. Syllabus**

| AI FOR TECHNOLOGY<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS308B** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/Sem (L:T: P:TW:SL) | **32:0:0:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course Objectives: The course aims**

1. To understand the need and significance of AI for technology.
2. To study advanced AI algorithms for technology.
3. Learn Computational Intelligence techniques.
4. To Understand evaluation metrics and ethics in intelligence for technical Systems.
5. To learn various NLP algorithms and their application in Healthcare and to investigate the current scope, implications of AI for developing futuristic technological Application.

**Module -1**

**Introduction:** Overview of AI, A Multifaceted Discipline, Application of AI in health care-prediction, diagnosis, personalized treatment and behaviour modification, drug discovery, follow-up care etc health Care-Use Cases.

**Textbook1: chapter1, chapter8** **10 Hours**

**Module -2**

**AI, Deep Learning and Data mining Methods for Health Care:** Knowledge discovery and data mining, Multi classifier decisions Fusion, Ensemble Learning, Meta-Learning and other Abstract Methods. Evolutionary algorithms, Illustrative Medical Application-Multiagent Infectious Disease propagation.

**Textbook2:chapter2, 3** **10 Hours**

**Module -3**

**Future of Health Care Using AI:** Evidence based medicine, Personalized Medicine, Block chain for verifying supply chain, patient record access, Robot-Assisted Surgery, smart hospitals, Conversational AI, patient record access, Robert-Assisted Surgery, Smart hospitals.

**Textbook1: chapter 7** **10 Hours**

**TEXTBOOKS:**

**1. A**rjun Panesar," Machine Learning and AI for Healthcare", A press.
**2.** Arvin Agah," Medical applications of Artificial Systems", CRC Press
**3.** SatyaRanjan Dash, ShantipriyaParida, Esau VillatoroTello, BiswaranjanAcharya, OndrejBojar "Natural Language processing in healthcare"

**REFERENCE BOOKS:**

**1**.Erik R. Ranschaert Sergey Morozov Paul R. Algra," Artificial Intelligence in medical Imaging-Opportunities, Applications and Risks", Springer.
**2.** Dac-Nhuong Le, Chung Van Le, Jolanda G. Tromp, Gia Nhu Nguyen," Emerging technologies for health and medicine", Wiley.
**3**.Ton J. CleophasAeilko H. Zwinderman," Machine Learning in Medicine-Complete Overview", Springer

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of AI concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |

4. **Assessment Details (both CIE and SEE)**

**Continuous Internal Evaluation:**

**CIE Split up for Ability Enhancement Course (AE)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B) | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks = (A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

1.The CIE question paper shall have MCQ set for 25 Questions. Each carrying one mark.

2.The average internal assessment shall be the average of the 2 best marks in the tests conducted.

**Semester End Examination:**

1. The SEE question paper shall have MCQ set for 50 questions, each carrying one mark.
2. The time duration for SEE is one hour.

**5.Course Outcomes (COs) and Mapping with POs/ PSOs**

**Course Outcomes (COs)**

| Cos | Description |
|---|---|
| **M25BCS308B.1** | **Understanding** the fundamental concepts of artificial intelligence and its applications in technology. |
| **M25BCS308B.2** | **Apply** AI technique to health care data for predictive modelling and decision-making. |
| **M25BCS308B.3** | **Analyze** the role machine learning & deep learning in medical diagnosis treatment. |
| **M25BCS308B.4** | **Evaluate** the ethical and legal implications of AI in technology. |
| **M25BCS308B.5** | **Design** and develop AI-powered health care system and solutions |

**CO-PO-PSO Mapping**

| COs/Pos | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS308.1** | 3 | - | - | - | - | - | - | - | - | - | - | 2 | 2 |
| **M25BCS308.2** | - | 3 | - | - | - | - | - | - | - | - | - | - | - |
| **M25BCS308.3** | - | - | 3 | 3 | - | - | - | - | - | - | - | 2 | 2 |
| **M25BCS308.4** | - | - | - | - | 3 | - | - | - | - | - | - | 2 | - |
| **M25BCS308.5** | **3** | **3** | **3** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **2** | **2** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 3**: Good Health and Well-Being | AI can improve health care outcomes, disease diagnosis, and treatment planning. |
| 2 | **SDG 9:** Industry, innovation**,** and | AI can drive innovation in health care technology, medical devices, and infrastructure.AI powered solutions can improve healthcare |

| | | |
|---|---|---|
| | infrastructure | accessibility and affordability. |
| 3 | **SDG 10**: Reduced Inequalities | AI can help reduce healthcare disparities by providing access to quality healthcare for underserved populations.<br>AI powered telemedicine can bridge the gap in healthcare access for rural and remote areas. |

| 3rd Semester | Ability Enhancement Course (AE)<br>DIGITAL FORENSICS | M25BCS308C |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Computer Fundamentals: | Basic knowledge of computer hardware, operating systems (windows & Linux), and file system. Understanding of system boot process, file structures, and storage devices. |
| 2 | Programming Knowledge: | Familiarity with basic programming concepts (C/C++/python). Ability to write simple scripts for data parsing and automation. |
| 3 | Networking Concepts: | Understanding of network protocols (TCP/IP, HTTP, DNS). Basic Knowledge of IP addressing, routing, and tools like wire shark. |
| 4 | Legal and Ethical Awareness: | Basic understanding of IT laws, data privacy regulations, and ethical considerations in cyber security and digital evidence handling. |

**2. Syllabus**

**DIGITIAL FORENSICS**
**SEMESTER – III**

| | | | |
|---|---|---|---|
| Course Code | **M25BCS308C** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem(L:T: P:TW:SL) | **32:0:0:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course Objectives:**

1. To understand the basic concepts of digital forensics and its role in investigating cybercrimes.
2. To identify and preserve digital evidence using proper forensic procedure.
3. To analyze data from computers, mobile devices, and networks using forensic tools and techniques.
4. To examine file systems, system logs, and deleted files to extract useful forensic evidence.
5. To study the legal and ethical issues related to digital investigations and cyber laws.
6. To gain hands-on experience in conducting forensic analysis using industry-standard tools.

**Module -1**

**Computer Forensics Fundamentals:** Introduction to Computer Forensic, Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps Taken by Computer Forensics Specialists, Use Computer Forensic Evidence.

**Text Book1: Chapter 1** **10 Hours**

**Module -2**

**Types of Computer Forensics Technology:** Types of Military Computer Forensic Technology, Types of Law Enforcement: Computer Forensic Technology, Types of Business Computer Forensic Technology, Specialized Forensics Techniques, Hidden Data, Spyware and Adware, Encryption Methods and Vulnerabilities, Protecting Data from Being Compromised, Internet Tracing Methods, Security and Wireless Technologies, Avoiding Pitfalls with Firewalls, Biometric Security Systems

Text Book1: Chapter 2 10 Hours

**Module -3**

**Types of Computer Forensics System:** Internet Security Systems, Intrusion Detection System, Firewall Security Systems, Storage Area Network Security Systems , Network Disaster Recovery Systems, Public Key Infrastructure Systems, Wireless Network Security Systems , Satellite Encryption Security Systems, Instant Messaging (IM) Security Systems, Net Privacy Systems, Identity Management Security Systems, Identity Theft

Text Book 1: Chapter 3 10 Hours

**TEXTBOOKS:**

1. Computer Forensics: Computer Crime Scene Investigation, 2nd Edition, John R. Vacca, Charles River Media, 2005.
2. Live Hacking: The Ultimate Guide to Hacking Techniques & Countermeasures for Ethical Hackers & IT Security Experts, Ali Jahangiri,First edition, 2009

**REFERENCE BOOKS:**

1. SunitBelapure and Nina Godbole. Cyber Security: Understanding Cyber Crimes, Computer Forensics And Legal Perspectives. Wiley India Pvt Ltd. 2013.

2. Understanding Cryptography: A Textbook for Students and Practitioners, ChristofPaar, Jan Pelzl, 2010, Second Edition, Springer's.
3. Live Hacking: The Ultimate Guide to Hacking Techniques & Countermeasures for Ethical Hackers & IT Security Experts, Ali Jahangiri,First edition, 2009.

**VIDEO LINKS:**

1. https://onlinecourses.nptel.ac.in/noc22_cs54/preview.
2. https://www.youtube.com/playlist?list=PL0sMmF6CJX4QG4gzrq3v49G-0BZpytqBR.

### 3. Teaching-Learning Process Strategies

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 6 | Laboratory Learning | Utilize the facilities available in the laboratories to understand the behavior of the materials by performing few experiments. |

### 4. Assessment Details (both CIE and SEE)

**Continuous Internal Evaluation:**

**CIE Split up for Ability Enhancement Course (AE)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B)* | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks =(A) + (B)**

**A** = Average of the best two Test marks

**B** = Average of two Term Work marks

***Assessment Plan**

- MCQ Quiz
- Case Study – Based MCQ Assignment
- Tool Awareness Task

1. The CIE question paper shall have MCQ set for 25 questions, each carrying one mark.
2. The average internal assessment shall be the average of the 2 best marks in the tests conducted

**Semester End Examination:**

1. The SEE question paper shall have MCQ set for 50 questions, each carrying one mark.
2. The time duration for SEE is one hour

### 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| Cos | Description |
|---|---|
| **M25BCS308C.1** | Understand and apply cyber forensics needs and types of Computer Forensics Systems. |
| **M25BCS308C.2** | Demonstrate the ability to utilize different types of computer forensics systems to investigate and analyse digital evidence effectively. |
| **M25BCS308C.3** | Analyse the steps involved in collecting digital evidence. |
| **M25BCS308C.4** | Analyse Design Cloud Storage Forensic. |
| **M25BCS308C.5** | Analyse the collection of evidence from cloud storage services. |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS308C.1** | 3 | - | - | - | - | - | - | - | - | - | - | 3 | - |
| **M25BCS308C.2** | - | - | - | - | - | - | - | - | - | - | - | 3 | 3 |
| **M25BCS308C.3** | - | - | 3 | 3 | - | - | - | - | - | - | - | 3 | 3 |
| **M25BCS308C.4** | - | - | 3 | 3 | - | - | - | - | - | - | - | 3 | 3 |
| **M25BCS308C.5** | - | - | - | 3 | - | - | - | - | - | - | - | 3 | - |
| **M25BCS308C** | **3** | **-** | **3** | **3** | **-** | **-** | **-** | **-** | **-** | **-** | **-** | **3** | **3** |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 4**: Quality Education | Provides students with technical skills in cyber investigation, evidence analysis, and ethical hacking, ensuring access to quality and relevant education.. |
| 2 | **SDG 9**: Industry , Innovation and Infrastructure | Digital forensics improves secure digital infrastructure and supports innovation in cyber security practices across industries. |
| 3 | **SDG 16**: Peace, Justice and Strong Institutions. | Promotes justice through lawful cyber practices and digital evidence investigation, contributing to strong and reliable justice institutions**.** |

| 3rd Semester | Ability Enhancement Course (AE)<br>DRUG DISCOVERY WITH AI | M25BCS308D |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Basic Biology | Understanding of cells, DNA, proteins, and biological pathways. Familiarity with drug-receptor interaction and basic pharmacokinetics (ADME). |
| 2 | Programming Basics | Proficiency in at least one programming language (preferably Python). Understanding of control flow, data structures, and libraries such as NumPy and Pandas. |
| 3 | Machine Learning | Familiarity with supervised and unsupervised learning, model evaluation, over fitting, and commonly used algorithms like decision trees and neural networks. |
| 4 | Data Handling & Statistics | Basic statistical knowledge (mean, standard deviation, correlation). Skills in data preprocessing, normalization, and visualization techniques. |

**2. Syllabus**

| DRUG DISCOVERY WITH AI<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BCS308D** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem(L:T: P:TW:SL) | **32:0:0:0:0 = 32 Hours** | SEE Marks | **50** |
| | | Total Marks | **100** |
| Credits | **01** | Exam Hours | **03** |

**Course Objectives:**

1. To understand the role of Artificial Intelligence in the drug discovery and development pipeline.
2. To apply machine learning techniques for target identification, drug screening, and optimization.
3. To gain knowledge of data preprocessing, molecular representations, and feature extraction methods.
4. To use AI tools for predicting drug-likeness, toxicity, and bioactivity.
5. To analyze biomedical datasets using AI for better decision-making in drug design.
6. To get familiar with software and computational platforms used in pharmaceutical AI applications.

**Module -1**

**Macromolecules and Their Structural Importance:** Central Dogma of molecular Biology, Structure of DNA, RNA, Amino acids, Classes of Protein architecture, Structure stabilizing interactions, protein folding problem, Solving protein structures, Mechanism of enzyme actions.

**Text Book: Chapter 1,2** **10 Hours**

**Module -2**

**Computer Representation for Developing 2D and 3D Models:** Computer representation of 2D chemical structures, Graph theory to represent Chemical Structures, Connection table, Computer representation of 3D chemical structures, biological databases.

**Text Book: Chapter 3,5** **10 Hours**

**Module -3**

**Drug Discovery Pipeline:** Sequence analysis, Methods of sequence analysis, Introduction to drug discovery process and computational approaches," Drug-Likeness" and Compound Filters, ligand and Structure based drug design, Virtual screening, Protein-Ligand Docking.

**Text Book: Chapter 4,6** **10 Hours**

**TEXTBOOKS:**

1. T. K. Attwood & D. J. Parry-Smith, Introduction to Bioinformatics, 1st Edition, Pearson Education, 2005.
2. AurélienGéron, Hands-On Machine Learning with Scikit-Learn, Keras&TensorFlow, 2nd Edition, O'Reilly Media, 2019.

**REFERENCE BOOKS:**

1. R. Mount, Bioinformatics: Sequence and Genome Analysis, 2nd Edition, Cold Spring Harbor Laboratory Press, 2004.
2. BharatiJaiswal, Artificial Intelligence in Drug Discovery and Development, CRC Press, 2022.

**VIDEO LINKS:**

1. https://onlinecourses.nptel.ac.in/noc22_bt38/preview
2. https://ocw.mit.edu/courses/7-91j-foundations-of-computational-and-systems-biology-spring-2014

### 3. Teaching-Learning Process Strategies

| Sl. No. | TLP Strategies | Description |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding of concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |
| 6 | Laboratory Learning | Utilize the facilities available in the laboratories to understand the behavior of the materials by performing few experiments. |

### 4. Assessment Details (both CIE and SEE)

**Continuous Internal Evaluation:**

**CIE Split up for Ability Enhancement Course (AE)**

| | Components | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| 1 | Internal Assessment-Tests (A) | 3 | 50% | 25 |
| 2 | Team Work (B) | 2 | 50% | 25 |
| | **Total Marks** | | | **50** |

**Final CIE Marks =(A) + (B)**

**A** = Average of best two Test marks

**B** = Average of two Term Work marks

*** Assessment Plan:**

- MCQ Quiz
- Case Study -Based MCQ Assignment
- Tool Awareness Task

1. The CIE question paper shall have MCQ set for 25 questions, each carrying one mark.
2. The average internal assessment shall be the average of the 2 best marks in the tests conducted

**Semester End Examination:**

1. The SEE question paper shall have MCQ set for 50 questions, each carrying one mark.
2. The time duration for SEE is one hour.

### 5. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BCS308D.1** | **Understand** knowledge about macromolecules and their structural importance. |
| **M25BCS308D.2** | **Apply** knowledge on developing 2D and 3D models. |
| **M25BCS308D.3** | **Build** knowledge about drug discovery pipeline. |

**CO-PO-PSO Mapping**

| COs/POs | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **M25BCS308D.1** | 2 | 2 | - | - | - | - | - | - | - | - | - | 2 | - |
| **M25BCS308D.2** | 2 | 3 | 2 | - | - | - | - | - | - | - | - | 2 | 2 |
| **M25BCS308D.3** | 3 | 3 | 2 | - | 2 | - | - | - | - | - | - | - | 2 |
| **M25BCS308D** | **2.3** | **2.6** | **2** | **-** | **2** | **-** | **-** | **-** | **-** | **-** | **-** | **2** | **2** |

### 6. Mapping to Sustainable Development Goals (SDG):

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 3**: Good Health and Well-Being | The course helps learners use AI to accelerate drug discovery and development, improving access to effective medicines. (SDG 3) |
| 2 | **SDG 4**: Quality Education | Equips students with interdisciplinary skills in biology, AI, and data science, promoting technical education relevant to healthcare advancement. (SDG 4) |
| 3 | **SDG 9**: Industry, Innovation and Infrastructure | Encourages use of AI for pharmaceutical innovation, supporting modern infrastructure in healthcare and biotechnology industries. (SDG 9) |

| 3rd Semester | Non-Credit Mandatory Course (NCMC)<br>NATIONAL SERVICE SCHEME (NSS) &<br>UNIVERSAL HUMAN VALUES | M25BSHK309 |
|---|---|---|

**7. Syllabus**

## NATIONAL SERVICE SCHEME (NSS) & UNIVERSAL HUMAN VALUES

**SEMESTER – III**

| | | | |
|---|---|---|---|
| Course Code | **M25BSHK309** | CIE Marks | **50** |
| Total Number of Teaching-Learning Hours/sem (L:T:P:TW:SL) | **0:0: 16: 0:0 = 16 Hours** | SEE Marks | **-** |
| | | Total Marks | **50** |
| Credits | **00** | Exam Hours | **-** |

**Course Objectives:**

1. Understand the community in general in which they work.
2. Identify the needs and problems of the community and involve them in problem –solving.
3. Develop among themselves a sense of social & civic responsibility & utilize their knowledgein finding practical solutions to individual and community problems.
4. Develop competence required for group-living and sharing of responsibilities & gain skills in mobilizing community participation to acquire leadership qualities and democratic attitudes.
5. Develop capacity to meet emergencies and natural disasters & practice national integration and social harmony in general.
6. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
7. To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of Human reality and the rest of existence. Such a holistic perspective forms the basis of Universal Human Values and the movement towards value-based living in a natural way.
8. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.
9. This course is intended to provide a much-needed orientation input in value education to the young enquiring minds.

**General Instructions - Pedagogy :**

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the activities will develop students' theoretical and applied social and cultural skills.
2. State the need for NSS activities and its present relevance in the society and Provide real-life examples.
3. Support and guide the students for self-planned activities.
4. You will also be responsible for assigning homework, grading assignments and quizzes, and documenting students' progress in real activities in the field.
5. Encourage the students for group work to improve their creative and analytical skills.
6. The methodology of this course is explorational and thus universally adaptable. It involves a systematic and rational study of the human being vis-à-vis the rest of existence.
7. In addition to the traditional lecture method, different types of innovative teaching methods may be adopted so that the activities will develop students' theoretical and applied skills.
8. State the need for UHV activities and their present relevance in society and provide real-life examples.
9. Support and guide the students in self-study activities.
10. You will also be responsible for assigning homework, grading assignments and quizzes, and documenting students' progress in real activities in the field.
11. This process of self-exploration takes the form of a dialogue between the teacher and the students to begin with, and then to continue within the student in every activity, leading to continuous self-evolution.
12. Encourage the students for group work to improve their creative and analytical skills.

## PART-A
## NATIONAL SERVICE SCHEME (NSS)

1. Organic farming, Indian Agriculture (Past, Present and Future) Connectivity for marketing.
2. Waste management– Public, Private and Govt organization, 5 R's.
3. Setting of the information imparting club for women leading to contribution in social and economic issues.

4. Water conservation techniques – Role of different stakeholders– Implementation.
5. Preparing an actionable business proposal for enhancing the village income and approach for implementation.
6. Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education.
7. Developing Sustainable Water management system for rural areas and implementation approaches.
8. Contribution to any national level initiative of Government of India. For eg. Digital India, Skill India, Swatch Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc.
9. Spreading public awareness under rural outreach programs.(minimum5 programs).
10. Social connect and responsibilities.
11. Plantation and adoption of plants. Know your plants.
12. Organize National integration and social harmony events /workshops /seminars. (Minimum 02 programs).
13. Govt. school Rejuvenation and helping them to achieve good infrastructure.

NOTE:

1. Student/s in individual or in a group Should select any one activity in the beginning of each semester till end of that respective semester for successful completion as per the instructions of NSS officer with the consent of HOD of the department.
2. At the end of every semester, activity report should be submitted for evaluation.

**Distribution of Activities**

1. Organic farming, Indian Agriculture (Past, Present and Future) Connectivity for marketing.
2. Waste management– Public, Private and Govt organization, 5 R's.
3. Setting of the information imparting club for women leading to contribution in social & economic issues.
4. Water conservation techniques – Role of different stakeholders– Implementation.
5. Preparing an actionable business proposal for enhancing the village income and approach for implementation.
6. Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education.
7. Developing Sustainable Water management system for rural areas and implementation approaches.
8. Contribution to any national level initiative of Government of India. For eg. Digital India, Skill India, Swatch Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc.
9. Spreading public awareness under rural outreach programs.(minimum5 programs).
10. Social connect and responsibilities.
11. Plantation and adoption of plants. Know your plants.
12. Organize National integration and social harmony events /workshops /seminars. (Minimum 02 programs).
13. Govt. school Rejuvenation and helping them to achieve good infrastructure.

## PART-B-Universal Human Values

**Part -1**

**Introduction to Value Education**

Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) Understanding Value Education, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Happiness and Prosperity – Current Scenario, Method to Fulfil the Basic Human Aspirations

**Part -2**

**Harmony in the Human Being:**

Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self, Understanding Harmony in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health

**Part -3**

**Harmony in the Family and Society:**

Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationships, 'Respect' – as the Right Evaluation, Other Feelings, Justice in Human-to-Human Relationships, Understanding Harmony in Society, Vision for the Universal Human Order

**Part -4**

**Harmony in the Nature/Existence:**

Understanding Harmony in Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the

Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence

**Part -5**

**Implications of the Holistic Understanding – a Look at Professional Ethics:**
Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession

**Text Book and Teachers Manual**

1. The Textbook A Foundation Course in Human Values and Professional Ethics, R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978- 93-87034- 47-1
2. The Teacher‟s Manual for A Foundation Course in Human Values and Professional Ethics, R R Gaur, R Asthana, G

**Reference Books**

1. JeevanVidya: EkParichaya, A Nagaraj, JeevanVidyaPrakashan, Amar kantak, 1999.
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).

**Web links and Video Lectures (e-Resources):**

1. https://www.uhv.org.in/uhv-ii
2. http://uhv.ac.in
3. http://www.storyofstuff.com
4. Al Gore, An Inconvenient Truth, Paramount Classics, USA Charlie Chaplin, Modern Times, United Artists, USA
5. IIT Delhi, Modern Technology – the Untold StoryGandhi A., Right Here Right Now, Cyclewala Productions
6. https://www.youtube.com/channel/UCQxWr5QB_eZUnwxSwxXEkQw
7. https://fdp-si.aicte-india.org/8dayUHV_download.php
8. https://www.youtube.com/watch?v=8ovkLRYXIjE
9. https://www.youtube.com/watch?v=OgdNx0X923I
10. https://www.youtube.com/watch?v=nGRcbRpvGoU
11. https://www.youtube.com/watch?v=sDxGXOgYEKM

## 8. Course Outcomes (COs) and Mapping with POs/ PSOs

**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BSHK309.1** | Explain the principles of value education, human aspirations, and holistic development in the context of individual and societal well-being. |
| **M25BSHK309.2** | Apply concepts of self-regulation, health, and harmony between the self and the body for personal growth and ethical living. |
| **M25BSHK309.3** | Demonstrate understanding of trust, respect, justice, and social responsibility to promote harmony within families and society. |
| **M25BSHK309.4** | Analyze the interdependence of humans and nature and propose sustainable solutions for environmental conservation and resource management. |
| **M25BSHK309.5** | Develop and implement community-oriented projects addressing rural development, education, entrepreneurship, and national initiatives. |
| **M25BSHK309.6** | Exhibit professional ethics and value-based decision-making in engineering practice for the realization of a humane and sustainable social order. |

**Pedagogy – Guidelines: NSS**

It may differ depending on local resources available for the study as well as environment and climatic differences, location, and time of execution.

| Sl No | Topic | Group size | Location | Activity execution | Reporting | Evaluation of the Topic |
|---|---|---|---|---|---|---|
| 1. | Organic farming, Indian Agriculture (Past, Present, and Future) Connectivity for marketing. | May be individual or team | Farmers land/ Villages/ roadside/ community area /College campus etc | Site selection / proper consultation/ Continuous monitoring/ Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 2. | Waste management– Public, Private and Govt organization, 5 R's. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc… | Site selection / proper consultation/ Continuous monitoring/ Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 3. | Setting of the information imparting club for women leading to contribution in social and economic issues. | May be individual or team | Women empowerment groups/ Consulting NGOs &Govt Teams / College campus etc… | Group selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 4. | Water conservation techniques – Role of different stakeholders– Implementation. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers, campus etc… | Site selection / Proper consultation/ Continuous monitoring/ Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 5. | Preparing an actionable business proposal for enhancing the village income and approach for implementation. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers, campus etc… | Group selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 6. | Helping local schools to achieve good results and enhance their enrolment in Higher/ technical/ vocational education. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers, campus etc… | School selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |

| | | | | | | |
|---|---|---|---|---|---|---|
| 7. | Developing Sustainable Water management system for rural areas and implementation approaches. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc… | Site selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 8. | Contribution to any national-level initiative of the Government of India. For eg. Digital India, Skill India, Swachh Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc… | Group selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 9. | Spreading public awareness under rural outreach programs. (minimum5 programs). Social connect and responsibilities. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc. | Group selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 10. | Plantation and adoption of plants. Know your plants. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc. | Place selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 11. | Organize National integration and social harmony events /workshops /seminars. (Minimum 02 programs). | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc… | Place selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |
| 12. | Govt. school Rejuvenation and helping them to achieve good infrastructure. | May be individual or team | Villages/ City Areas / Grama panchayat/ public associations/ Government Schemes officers / campus etc… | Place selection / proper consultation / Continuous monitoring / Information board | Report should be submitted by an individual to the concerned evaluation authority | Evaluation as per the rubrics of the scheme and syllabus by NSS officer |

**Plan of Action: NSS**

| Sl .No | Practice Session Description |
|---|---|
| 1 | Lecture session by NSS Officer |
| 2 | Students' Presentation on Topics |

| | |
|---|---|
| 3 | Presentation -1, Selection of topic, PHASE - 1 |
| 4 | Commencement of activity and its progress - PHASE - 2 |
| 5 | Execution of Activity |
| 6 | Execution of Activity |
| 7 | Execution of Activity |
| 8 | Execution of Activity |
| 9 | Execution of Activity |
| 10 | Case study-based Assessment, Individual performance |
| 11 | Sector-wise study and its consolidation |
| 12 | Video-based seminar for 10 minutes by each student. At the end of the semester with a Report. |

**Assessment Details:**

| **Weightage** | **CIE – 100%** | |
|---|---|---|
| Presentation – 1 - Selection of topic, PHASE - 1 | **10 Marks** | • Implementation strategies of the project (NSS work).<br>• The last Report should be signed by the NSS Officer, the HOD, and the principal.<br>• At last Report should be evaluated by the NSS officer of the institute.<br>• Finally, the consolidated marks sheet should be sent to the university and made available at the LIC visit. |
| Commencement of activity& its progress - PHASE - 2 | **10 Marks** | |
| Case Study-based Assessment Individual Performance with Report | **10 Marks** | |
| Sector-wise study & its consolidation | **10 Marks** | |
| Video based seminar for 10 minutes by each student. At the end of semester with Report. Activities. | **10 Marks** | |
| Total marks for the course. | **50 Marks** | |

**Suggested Learning Resources**:/**Books** :

1. NSS Course Manual, Published by NSS Cell, VTU Belagavi.
2. Government of Karnataka, NSS cell, activities reports and manual.
3. Government of India, NSS cell, Activities reports and manual.

**9. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | SDG 2 (Zero Hunger) | Promotes sustainable agriculture. |
| 2 | SDG 4 (Quality Education) | Enhances educational infrastructure. |
| 3 | SDG 5 (Gender Equality) | Directly supports gender-inclusive growth. |
| 4 | SDG 6 (Clean Water) | Implements rural water management systems. |
| 5 | SDG 8 (Decent Work) | Directly supports gender-inclusive growth. |
| 6 | SDG 11 (Sustainable Cities), | Addresses waste reduction and circular economy. |
| 7 | SDG 12 (Responsible Consumption) | |
| 8 | SDG 13 (Climate Action) | Emergency response aligns with climate resilience. |
| 9 | SDG 16 (Peace & Justice) | Fosters social harmony but indirectly linked. |

**Evaluation Guidelines**

**CIE Split up**

| **Components** | | **Number** | **Weightage** | **Max. Marks** |
|---|---|---|---|---|
| PART-A | | | | |
| (i) | NSS Activity-Report (A) | 1 | 50% | 25 |
| (ii) | Internal Assessment-UHV-Tests (B) | 3 | 50% | 25 |
| | **Total Marks (A+B)** | | | **50** |

1. The CIE question paper for Part-B (Universal Human Values) shall have MCQ set for 25 questions, each carrying one mark.
2. The average internal assessment shall be the average of the 2 best marks in the tests conducted

| 3rd Semester | Basic Science Course (BS)<br>ADDITIONAL MATHEMATICS - I | M25BDIPM310 |
|---|---|---|

**1. Prerequisites**

| Sl. No. | Proficiency | Pre-requisites |
|---|---|---|
| 1 | Calculus: | Knowledge of calculus, specifically Successive differentiation-problems |
| 2 | Basic Concepts of Complex Numbers: | Strong knowledge of Complex Numbers, to solve differential equations. |
| 3 | Basic Mathematics: | Knowledge of advanced calculus, and ordinary differential equations Familiarity with identify the dependent and independent variables |
| 4 | Basic Concept of Vector Algebra and Vector Differentiation: | Knowledge of basics, solving Computer Scienceengineering problemsusing vector calculus. |

**2. Syllabus**

| ADDITIONAL MATHEMATICS - I<br>SEMESTER – III | | | |
|---|---|---|---|
| Course Code | **M25BDIPM310** | CIE Marks | **100** |
| Total Number of Teaching-Learning Hours/Semester (L:T: P:S:TW:SL) | **32: 0: 0: 0: 0:0=32 Hours** | SEE Marks | **-** |
| | | Total Marks | **100** |
| Credits | **0** | Exam Hours | **00** |

**Course Objectives:**

1. To learn basic operations on vector calculus.
2. To analyzing Successive differentiation.
3. To learn Basic concept of Complex Numbers.
4. Understanding of basic solution of algebraic and transcendental equations

**Module -1**

**Differential Calculus:** Successive differentiation-problems. Taylor's & Maclaurin's series, Expansions-problems. Partial Differentiation: Euler's theorem (without Proof)-problems on first order derivatives only. Total derivatives-differentiation of composite functions. Jacobians of order two-Problems.

**Module -2**

**Complex Numbers:** Definitions and properties. Modulus and amplitude of a complex number, Argand's diagram, De-Moivre's theorem (without proof), Problems. Vector Algebra: Scalar and vectors. Addition, subtraction and multiplication of vectors- Dot and Cross products, problems. Scalar triple product, Problems.

**Module -3**

**Vector Differentiation:** Differentiation of vector functions. Velocity and acceleration of a particle moving on a space curve. Scalar and vector point functions. Gradient, Divergence, Curl-simple problems. Solenoidal and irrotational vector fields-Problems.

**Module -4**

**Integral Calculus:**Review of elementary integral calculus. Reduction formulae for sinx, $\cos^n x$, $\sin^n x \cos^n x$ (without proof) and evaluation of these with standard limits-problems. Double and triple integrals-Simple problems.

**Module -5**

**Ordinary Differential Equations:** Introduction-solutions of first order and first-degree differential equations: Variable separable method, Homogeneous differential equations, linear differential equations. Exact differential equations.

**TEXTBOOKS:**

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 45th Edition, 2025.
2. Advanced Engineering Mathematics, E. Kreyszig John, Wiley & Sons, 10th Edition, 2015.

**3. Teaching-Learning Process Strategies**

| Sl. No. | TLP Strategies | Description |
|---|---|---|

| | | |
|---|---|---|
| 1 | Lecture Method | Utilize various teaching methods within the lecture format to reinforce competencies. |
| 2 | Video/Animation | Incorporate visual aids like videos/animations to enhance understanding the concepts. |
| 3 | Collaborative Learning | Encourage collaborative learning for improved competency application. |
| 4 | Real-World Application | Discuss practical applications to connect theoretical concepts with real-world competencies. |
| 5 | Flipped Class Technique | Utilize a flipped class approach, providing materials before class to facilitate deeper understanding of competencies |

**4. Assessment Details (both CIE and SEE)**

Continuous Internal Evaluation:

| Components | | Number | Weightage | Max. Marks |
|---|---|---|---|---|
| (i) | Internal Assessment - Tests (A) | 3 | 50% | 50 |
| (ii) | Term Work - TW (B) | 2 | 50% | 50 |
| | **Total Marks** | | | **100** |

**Final CIE Marks = (A) + (B)**
**A** = Average of best two Test marks
**B** = Average of two Term Work marks

**5. Course Outcomes (COs) and Mapping with POs/ PSOs**
**Course Outcomes (COs)**

| COs | Description |
|---|---|
| **M25BDIPM310.1** | Use derivatives and partial derivatives to calculate the rate of change of multivariate functions. |
| **M25BDIPM310.2** | Apply concepts of complex numbers and vector algebra to analyse the problems arising in a related area. |
| **M25BDIPM310.3** | Analyse position, velocity and acceleration in two and three dimensions of vector-valued functions. |
| **M25BDIPM310.4** | Learn techniques of integration including the evaluation of double and triple integrals. |
| **M25BDIPM310.5** | Identify and solve first-order ordinary differential equations. |

**6. Mapping to Sustainable Development Goals (SDG):**

| Sl. No. | SDG | Justification |
|---|---|---|
| 1 | **SDG 1:** No Poverty | Use first derivatives to model the rate of change in income or poverty levels over time. This helps in evaluating the effectiveness of poverty reduction programs. |
| 2 | **SDG 3:** Good Health and Well-being | Complex numbers can be used in epidemic modelling and signal processing of health data. Electrical signals in the human body (like EEG or ECG) are often analysed using complex impedance and Fourier transforms. |
| 3 | **SDG 7**: Affordable and Clean Energy | The rate of change of energy consumption can be modelled as a function of population growth, economic development, and energy efficiency improvements. Similarly, the rate of change in $CO_2$ emissions can be studied to evaluate the impact of clean energy policies. |